

9702/22/M/J/09/Q2

- 1 An experiment is conducted on the surface of the planet Mars. A sphere of mass 0.78 kg is projected almost vertically upwards from the surface of the planet. The variation with time t of the vertical velocity v in the upward direction is shown in Fig. 2.1.

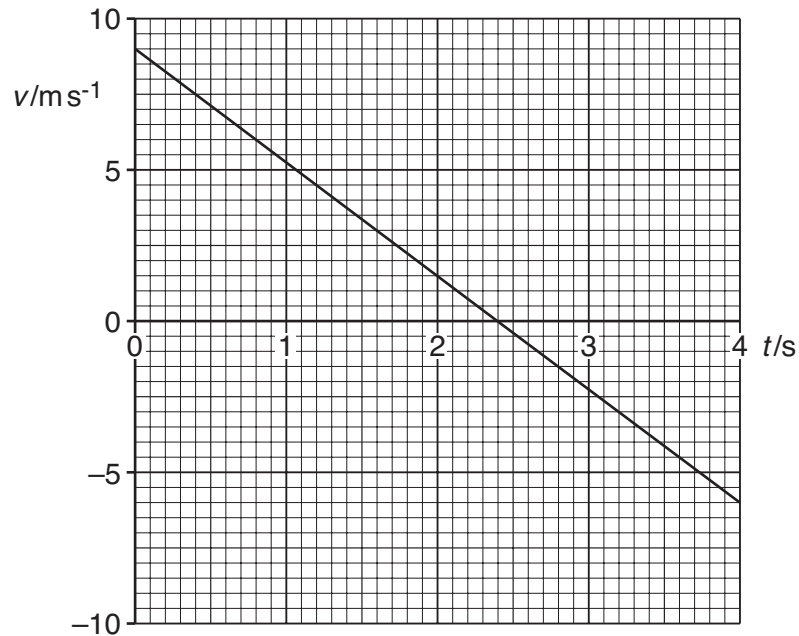


Fig. 2.1

The sphere lands on a small hill at time $t = 4.0$ s.

- (a) State the time t at which the sphere reaches its maximum height above the planet's surface.

$t = \dots\dots\dots$ s [1]

- (b) Determine the vertical height above the point of projection at which the sphere finally comes to rest on the hill.

height = $\dots\dots\dots$ m [3]

2 A sky-diver jumps from a high-altitude balloon.

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(a) Explain briefly why the acceleration of the sky-diver

(i) decreases with time,

.....

 [2]

(ii) is 9.8 ms^{-2} at the start of the jump.

.....
 [1]

(b) The variation with time t of the vertical speed v of the sky-diver is shown in Fig. 2.1.

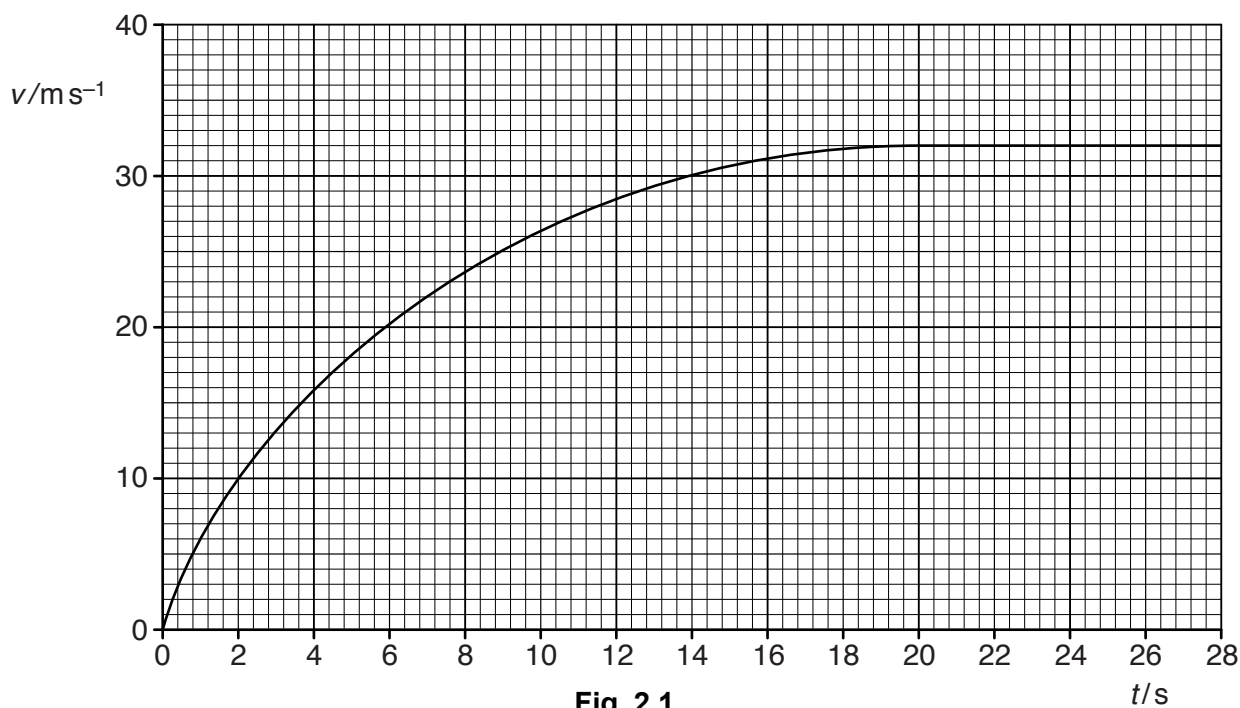


Fig. 2.1

Use Fig. 2.1 to determine the magnitude of the acceleration of the sky-diver at time $t = 6.0 \text{ s}$.

acceleration = m s^{-2} [3]

(c) The sky-diver and his equipment have a total mass of 90 kg.

(i) Calculate, for the sky-diver and his equipment,

1. the total weight,

weight = N [1]

2. the accelerating force at time $t = 6.0$ s.

force = N [1]

(ii) Use your answers in (i) to determine the total resistive force acting on the sky-diver at time $t = 6.0$ s.

force = N [1]

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- 3 A small ball is thrown horizontally with a speed of 4.0 m s^{-1} . It falls through a vertical height of 1.96 m before bouncing off a horizontal plate, as illustrated in Fig. 3.1.

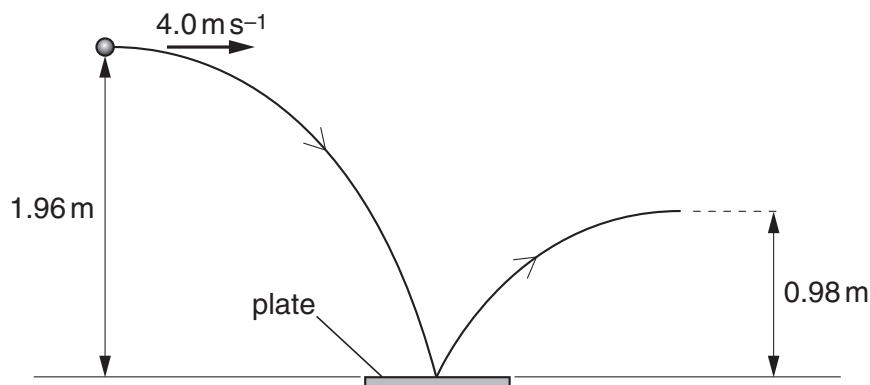


Fig. 3.1

Air resistance is negligible.

- (a) For the ball, as it hits the horizontal plate,
 (i) state the magnitude of the horizontal component of its velocity,

horizontal velocity = m s^{-1} [1]

- (ii) show that the vertical component of the velocity is 6.2 m s^{-1} .

[1]

- (b) The components of the velocity in (a) are both vectors.
 Complete Fig. 3.2 to draw a vector diagram, to scale, to determine the velocity of the ball as it hits the horizontal plate.

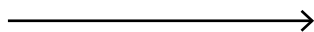


Fig. 3.2

velocity = m s^{-1}

at $^{\circ}$ to the vertical [3]

- (c) After bouncing on the plate, the ball rises to a vertical height of 0.98 m.
(i) Calculate the vertical component of the velocity of the ball as it leaves the plate.

vertical velocity = m s^{-1} [2]

- (ii) The ball of mass 34 g is in contact with the plate for a time of 0.12 s.
Use your answer in (c)(i) and the data in (a)(ii) to calculate, for the ball as it bounces on the plate,

1. the change in momentum,

change = kg m s^{-1} [3]

2. the magnitude of the average force exerted by the plate on the ball due to this momentum change.

force = N [2]

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- 4 (a) Complete Fig. 2.1 to show whether each of the quantities listed is a vector or a scalar.

	vector / scalar
distance moved	
speed	
acceleration	

Fig. 2.1

[3]

- (b) A ball falls vertically in air from rest. The variation with time t of the distance d moved by the ball is shown in Fig. 2.2.

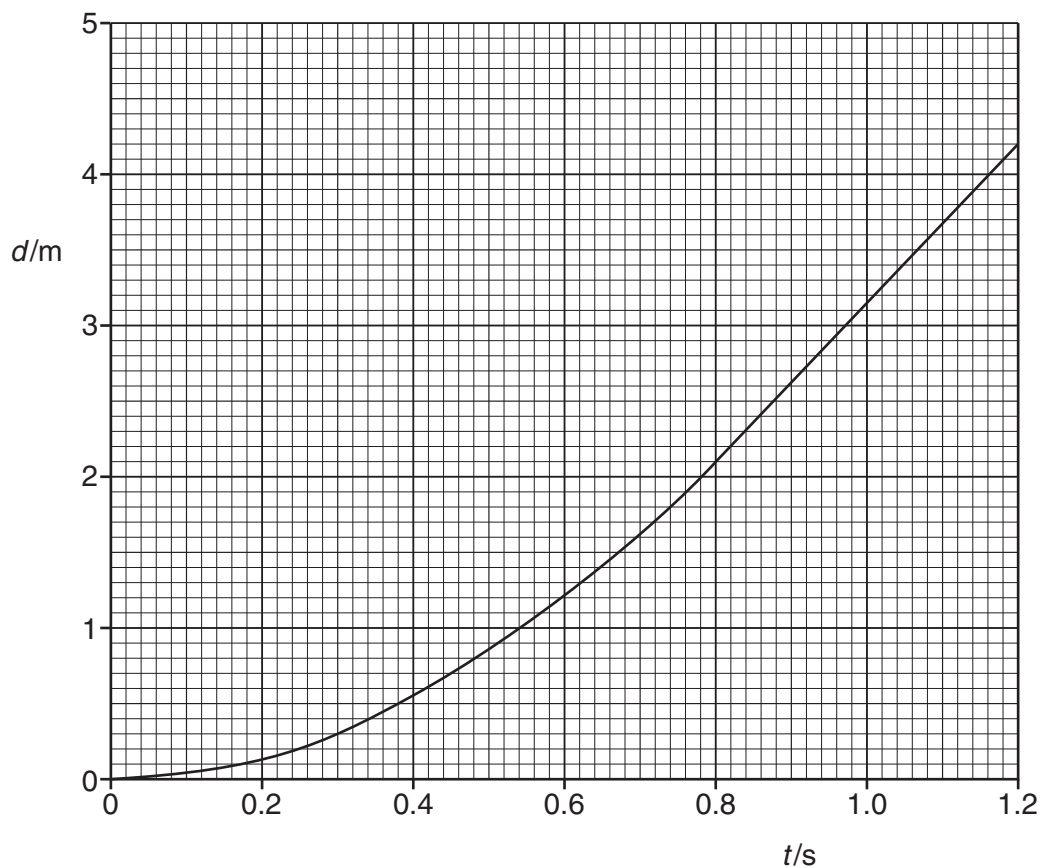


Fig. 2.2

- (i) By reference to Fig. 2.2, explain how it can be deduced that

1. the ball is initially at rest,

.....

 [2]

2. air resistance is not negligible.

.....
.....[1]

- (ii) Use Fig. 2.2 to determine the speed of the ball at a time of 0.40 s after it has been released.

speed = m s^{-1} [2]

- (iii) On Fig. 2.2, sketch a graph to show the variation with time t of the distance d moved by the ball for negligible air resistance. You are not expected to carry out any further calculations.

[3]

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- 5 (a)** The distance s moved by an object in time t may be given by the expression

$$s = \frac{1}{2}at^2$$

where a is the acceleration of the object.

State two conditions for this expression to apply to the motion of the object.

1.

.....

2.

.....

[2]

- (b)** A student takes a photograph of a steel ball of radius 5.0 cm as it falls from rest. The image of the ball is blurred, as illustrated in Fig. 2.1. The image is blurred because the ball is moving while the photograph is being taken.

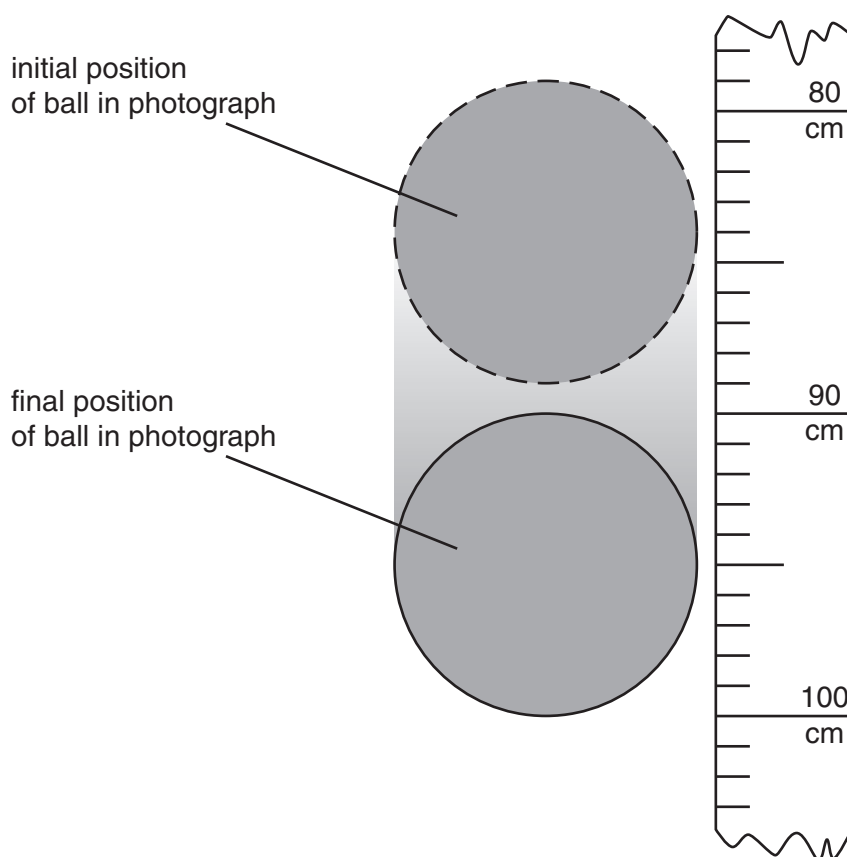


Fig. 2.1

The scale shows the distance fallen from rest by the ball. At time $t = 0$, the top of the ball is level with the zero mark on the scale. Air resistance is negligible.

Calculate, to an appropriate number of significant figures,

- (i) the time the ball falls before the photograph is taken,

time = s [3]

- (ii) the time interval during which the photograph is taken.

time interval = s [3]

- (c) The student in (b) takes a second photograph starting at the same position on the scale. The ball has the same radius but is less dense, so that air resistance is not negligible.

State and explain the changes that will occur in the photograph.

.....
.....
.....
..... [2]

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- 6 A ball is thrown horizontally from the top of a building, as shown in Fig. 2.1.

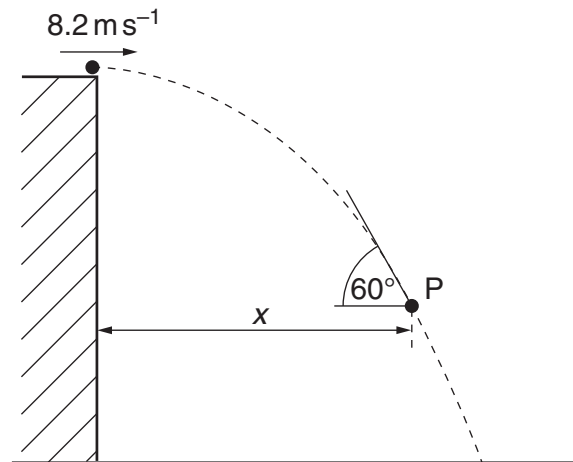


Fig. 2.1

The ball is thrown with a horizontal speed of 8.2 ms^{-1} . The side of the building is vertical. At point P on the path of the ball, the ball is distance x from the building and is moving at an angle of 60° to the horizontal. Air resistance is negligible.

- (a) For the ball at point P,

- (i) show that the vertical component of its velocity is 14.2 ms^{-1} ,

[2]

- (ii) determine the vertical distance through which the ball has fallen,

distance = m [2]

- (iii) determine the horizontal distance x .

$x = \dots\dots\dots$ m [2]

- (b) The path of the ball in (a), with an initial horizontal speed of 8.2 m s^{-1} , is shown again in Fig. 2.2.

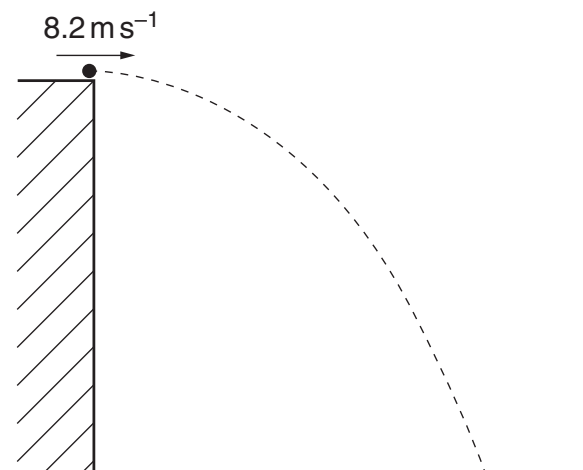


Fig. 2.2

On Fig. 2.2, sketch the new path of the ball for the ball having an initial horizontal speed

- (i) greater than 8.2 m s^{-1} and with negligible air resistance (label this path G), [2]
- (ii) equal to 8.2 m s^{-1} but with air resistance (label this path A). [2]
-

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- 7 A ball is thrown from a point P, which is at ground level, as illustrated in Fig. 2.1.

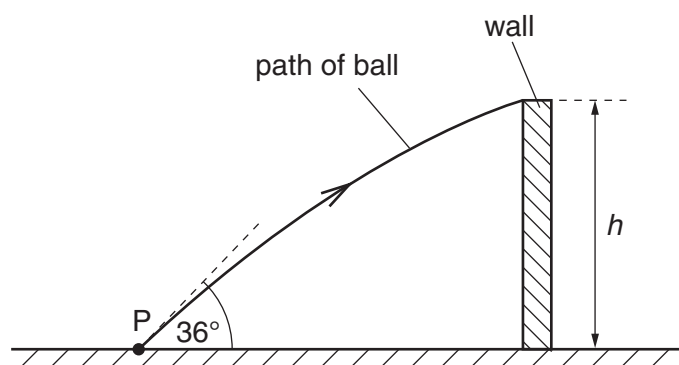


Fig. 2.1

The initial velocity of the ball is 12.4 m s^{-1} at an angle of 36° to the horizontal. The ball just passes over a wall of height h . The ball reaches the wall 0.17 s after it has been thrown.

- (a) Assuming air resistance to be negligible, calculate
- (i) the horizontal distance of point P from the wall,

distance = m [2]

- (ii) the height h of the wall.

$h = \dots\dots\dots \text{ m [3]}$

- (b) A second ball is thrown from point P with the same velocity as the ball in (a). For this ball, air resistance is not negligible. This ball hits the wall and rebounds.

On Fig. 2.1, sketch the path of this ball between point P and the point where it first hits the ground.

[2]

9702/22/M/J/11/Q1(c)

- 8 A stone is thrown with a horizontal velocity of 20 m s^{-1} from the top of a cliff 15 m high. The path of the stone is shown in Fig. 1.1.

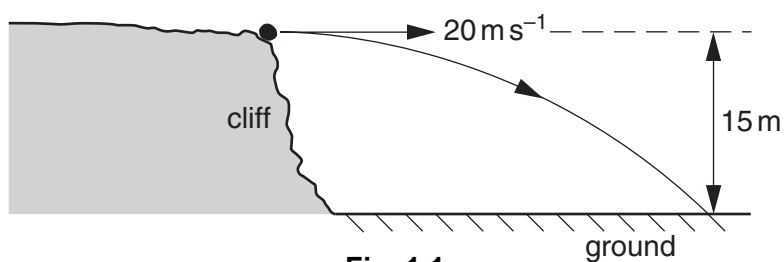


Fig. 1.1

Air resistance is negligible.

For this stone,

- (i) calculate the time to fall 15 m ,

time = s [2]

- (ii) calculate the magnitude of the resultant velocity after falling 15 m ,

resultant velocity = m s^{-1} [3]

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- 9 (a) A sphere of radius R is moving through a fluid with constant speed v . There is a frictional force F acting on the sphere, which is given by the expression

$$F = 6\pi DRv$$

where D depends on the fluid.

- (i) Show that the SI base units of the quantity D are $\text{kg m}^{-1} \text{s}^{-1}$.

[3]

- (ii) A raindrop of radius 1.5 mm falls vertically in air at a velocity of 3.7 m s^{-1} . The value of D for air is $6.6 \times 10^{-4} \text{ kg m}^{-1} \text{s}^{-1}$. The density of water is 1000 kg m^{-3} .

Calculate

1. the magnitude of the frictional force F ,

$$F = \dots\dots\dots \text{ N [1]}$$

2. the acceleration of the raindrop.

$$\text{acceleration} = \dots\dots\dots \text{ ms}^{-2} [3]$$

(b) The variation with time t of the speed v of the raindrop in (a) is shown in Fig. 2.1.

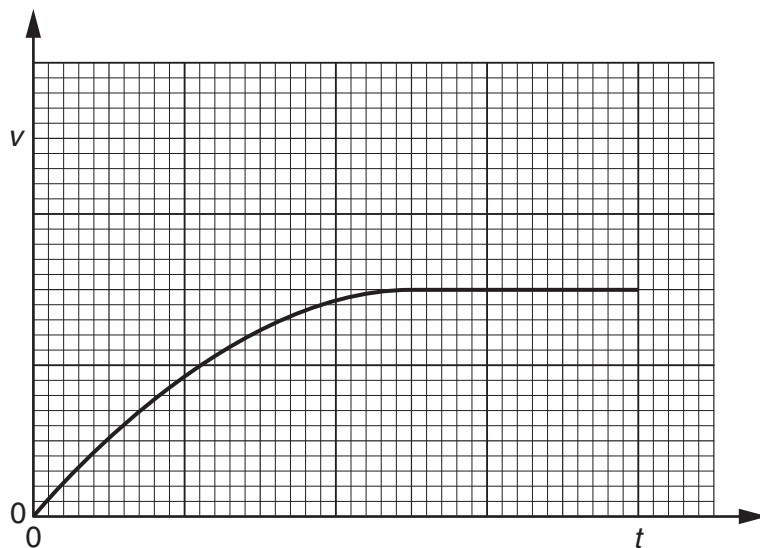


Fig. 2.1

(i) State the variation with time of the **acceleration** of the raindrop.

.....

.....

.....

.....

..... [3]

(ii) A second raindrop has a radius that is smaller than that given in (a). On Fig. 2.1, sketch the variation of speed with time for this second raindrop.

[2]

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- 10** A ball is thrown against a vertical wall. The path of the ball is shown in Fig. 3.1.

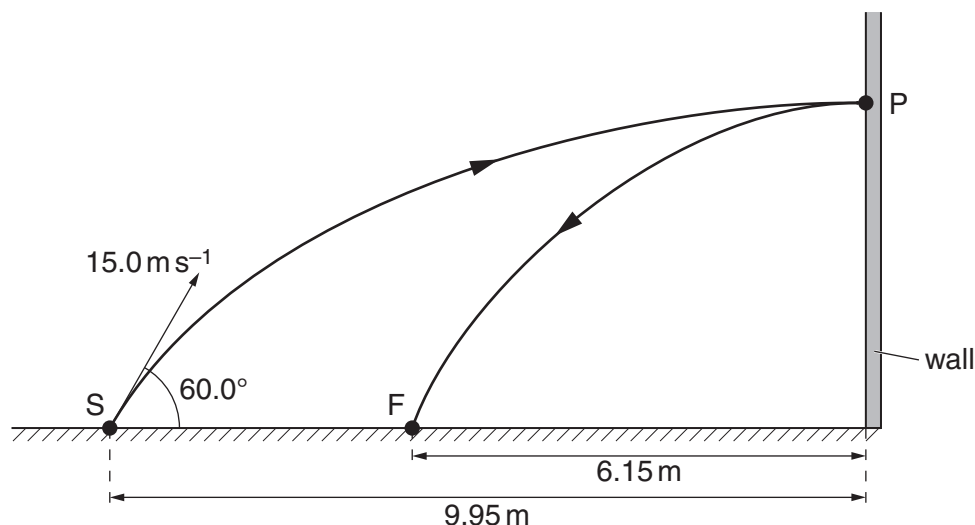


Fig. 3.1 (not to scale)

The ball is thrown from S with an initial velocity of 15.0 ms^{-1} at 60.0° to the horizontal. Assume that air resistance is negligible.

- (a)** For the ball at S, calculate

- (i)** its horizontal component of velocity,

horizontal component of velocity = ms^{-1} [1]

- (ii)** its vertical component of velocity.

vertical component of velocity = ms^{-1} [1]

- (b)** The horizontal distance from S to the wall is 9.95 m . The ball hits the wall at P with a velocity that is at right angles to the wall. The ball rebounds to a point F that is 6.15 m from the wall.

Using your answers in **(a)**,

- (i)** calculate the vertical height gained by the ball when it travels from S to P,

height = m [1]

- (ii) show that the time taken for the ball to travel from S to P is 1.33s,

[1]

- (iii) show that the velocity of the ball immediately after rebounding from the wall is about 4.6 m s^{-1} .

[1]

- (c) The mass of the ball is $60 \times 10^{-3} \text{ kg}$.

- (i) Calculate the change in momentum of the ball as it rebounds from the wall.

change in momentum = N s [2]

- (ii) State and explain whether the collision is elastic or inelastic.

.....

.....

..... [1]

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- 11** The variation with time t of the displacement s for a car is shown in Fig. 1.1.

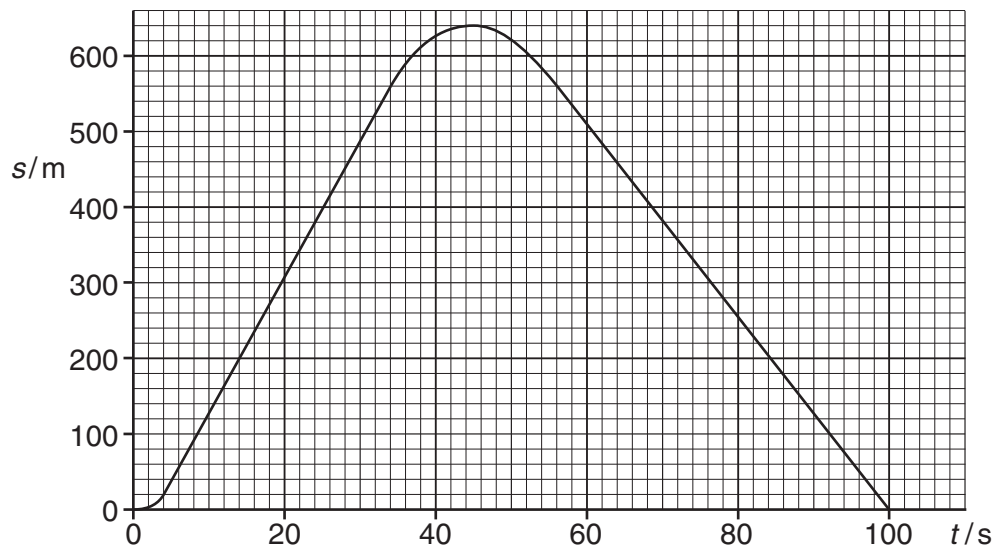


Fig. 1.1

- (a) Determine the magnitude of the average velocity between the times 5.0 s and 35.0 s.

average velocity = ms^{-1} [2]

- (b) On Fig. 1.2, sketch the variation with time t of the velocity v for the car.

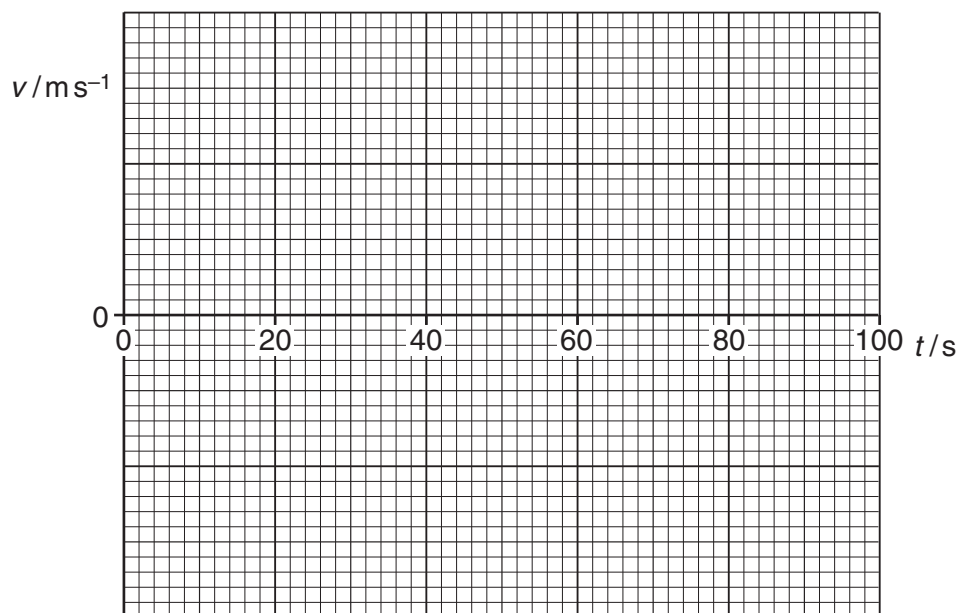


Fig. 1.2

[4]

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- 12** A ball is thrown vertically down towards the ground with an initial velocity of 4.23 m s^{-1} . The ball falls for a time of 1.51 s before hitting the ground. Air resistance is negligible.

(a) (i) Show that the downwards velocity of the ball when it hits the ground is 19.0 m s^{-1} .

[2]

(ii) Calculate, to three significant figures, the distance the ball falls to the ground.

distance = m [2]

- (b) The ball makes contact with the ground for 12.5 ms and rebounds with an upwards velocity of 18.6 m s^{-1} . The mass of the ball is 46.5 g.

(i) Calculate the average force acting on the ball on impact with the ground.

magnitude of force = N

direction of force

[4]

- (ii) Use conservation of energy to determine the maximum height the ball reaches after it hits the ground.

height = m [2]

- (c) State and explain whether the collision the ball makes with the ground is elastic or inelastic.

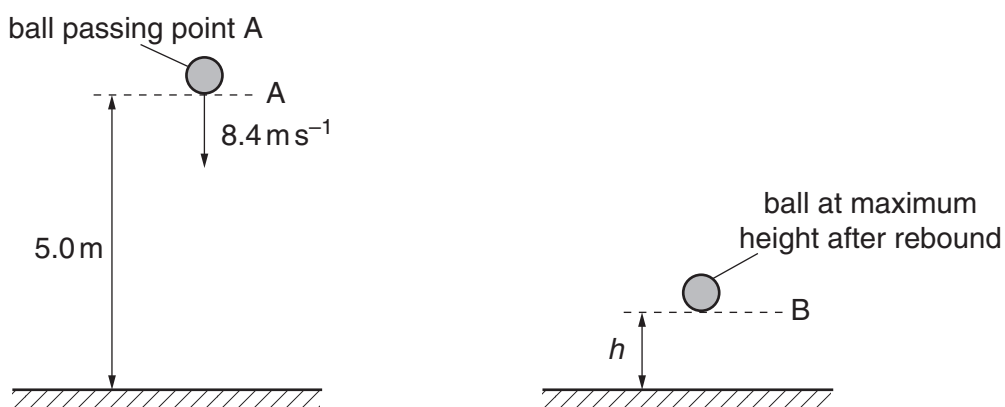
.....

.....

..... [1]

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- 13 (a)** A ball is thrown vertically down towards the ground and rebounds as illustrated in Fig. 2.1.

**Fig. 2.1**

As the ball passes A, it has a speed of 8.4 ms⁻¹. The height of A is 5.0 m above the ground. The ball hits the ground and rebounds to B. Assume that air resistance is negligible.

- (i) Calculate the speed of the ball as it hits the ground.

speed = ms⁻¹ [2]

- (ii) Show that the time taken for the ball to reach the ground is 0.47 s.

[1]

- (b) The ball rebounds vertically with a speed of 4.2 ms^{-1} as it leaves the ground. The time the ball is in contact with the ground is 20 ms . The ball rebounds to a maximum height h .

The ball passes A at time $t = 0$. On Fig. 2.2, plot a graph to show the variation with time t of the velocity v of the ball. Continue the graph until the ball has rebounded from the ground and reaches B.

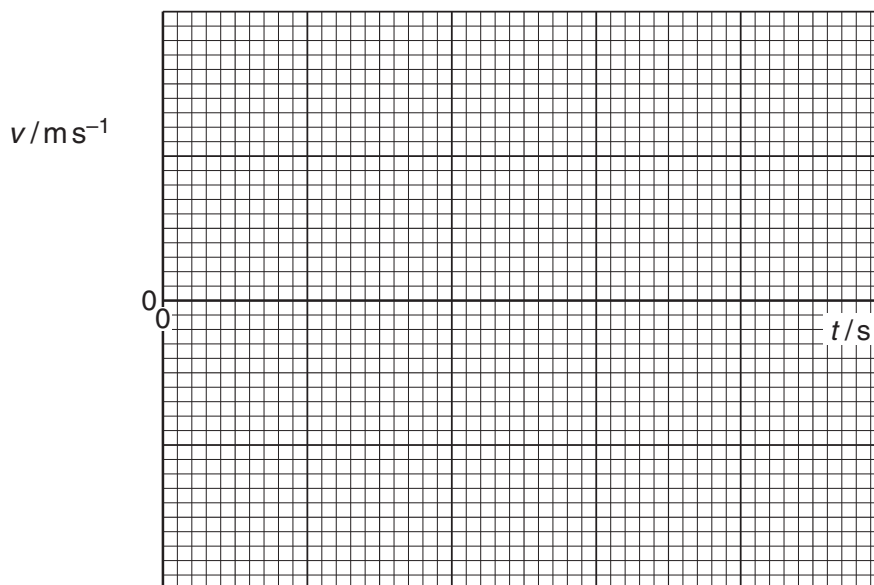


Fig. 2.2

[3]

- (c) The ball has a mass of 0.050 kg . It moves from A and reaches B after rebounding.

- (i) For this motion, calculate the change in
1. kinetic energy,

change in kinetic energy = J [2]

2. gravitational potential energy.

change in potential energy = J [3]

- (ii) State and explain the total change in energy of the ball for this motion.

.....

 [2]

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- 14 (a) Explain the differences between the quantities *distance* and *displacement*.

.....

[2]

- (b) State Newton's first law.

.....

[1]

- (c) Two tugs pull a tanker at constant velocity in the direction XY, as represented in Fig. 1.1.

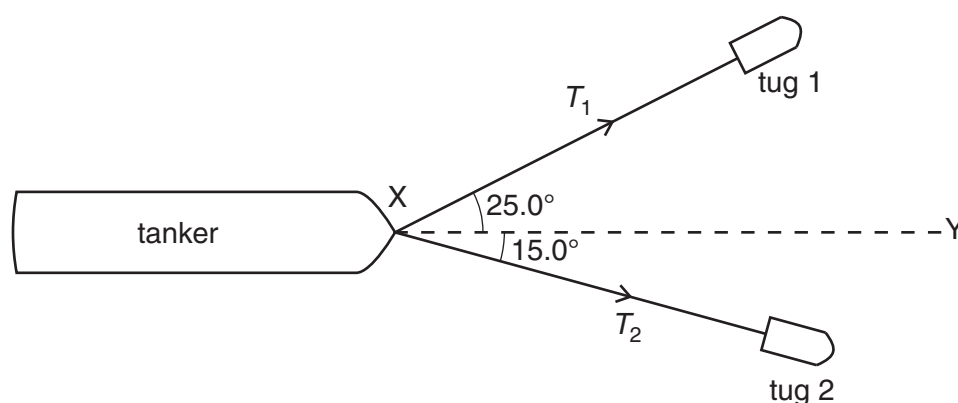


Fig. 1.1

Tug 1 pulls the tanker with a force T_1 at 25.0° to XY. Tug 2 pulls the tanker with a force of T_2 at 15.0° to XY. The resultant force R due to the two tugs is $25.0 \times 10^3 \text{ N}$ in the direction XY.

- (i) By reference to the forces acting on the tanker, explain how the tanker may be described as being in equilibrium.

.....

[2]

- (ii) 1. Complete Fig. 1.2 to draw a vector triangle for the forces R , T_1 and T_2 . [2]

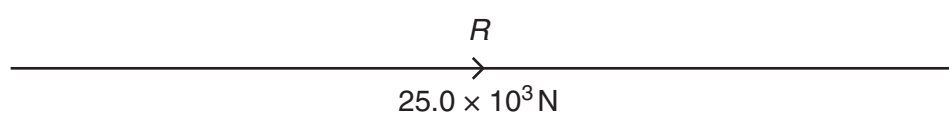


Fig. 1.2

2. Use your vector triangle in Fig. 1.2 to determine the magnitude of T_1 and of T_2 .

$$T_1 = \dots\dots\dots \text{ N}$$

$$T_2 = \dots\dots\dots \text{ N}$$

[2]

15 (a) (i) Define *acceleration*.

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.....
..... [1]

(ii) State Newton's first law of motion.

.....
..... [1]

(b) The variation with time t of vertical speed v of a parachutist falling from an aircraft is shown in Fig. 1.1.

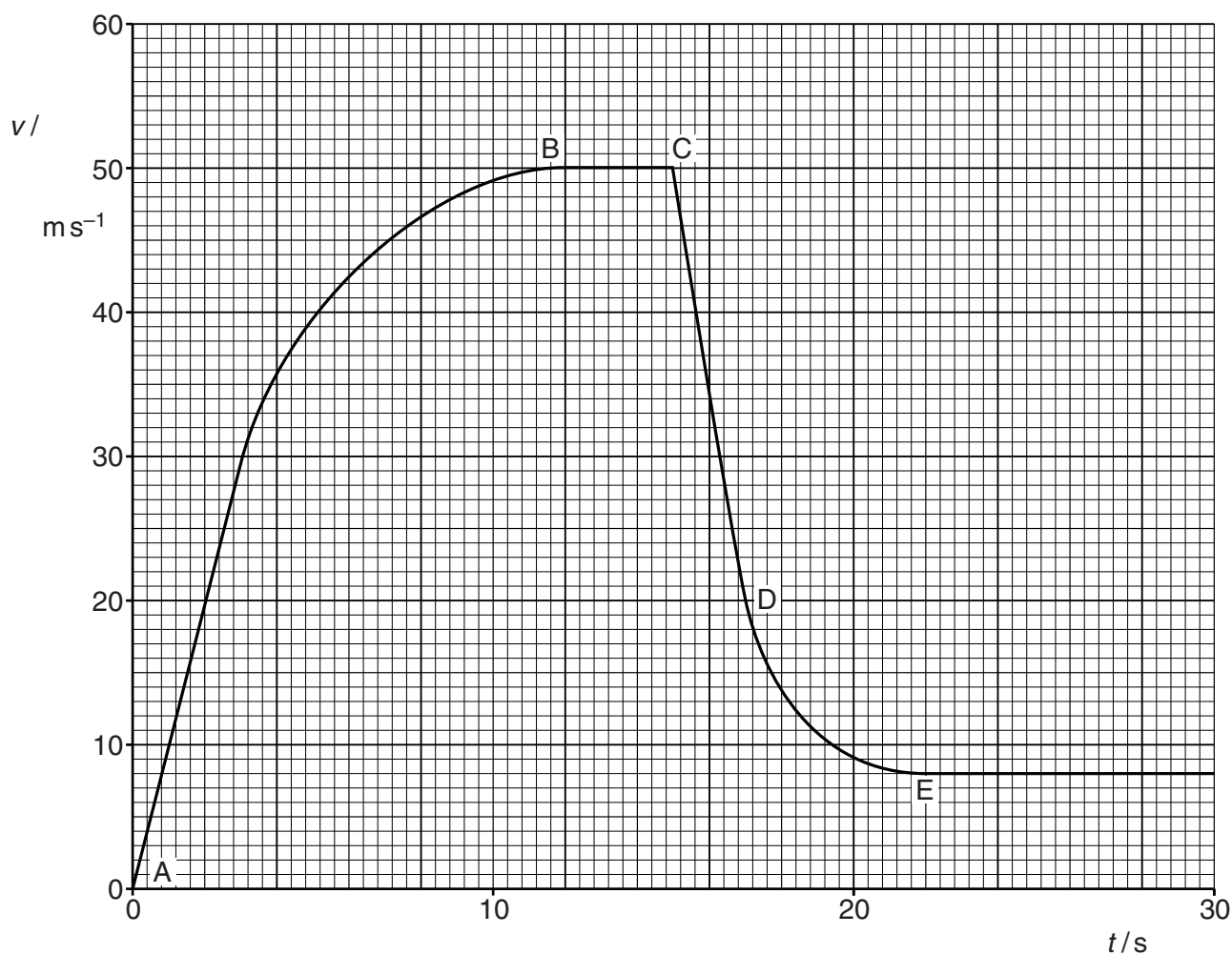


Fig. 1.1

- (i) Calculate the distance travelled by the parachutist in the first 3.0s of the motion.

distance = m [2]

- (ii) Explain the variation of the resultant force acting on the parachutist from $t = 0$ (point A) to $t = 15$ s (point C).

.....

 [3]

- (iii) Describe the changes to the frictional force on the parachutist

1. at $t = 15$ s (point C),

.....
 [1]

2. between $t = 15$ s (point C) and $t = 22$ s (point E).

.....
 [1]

- (iv) The mass of the parachutist is 95 kg.

Calculate, for the parachutist between $t = 15$ s (point C) and $t = 17$ s (point D),

1. the average acceleration,

acceleration = ms^{-2} [2]

2. the average frictional force.

frictional force = N [3]

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- 16 (a)** The drag force D on an object of cross-sectional area A , moving with a speed v through a fluid of density ρ , is given by

$$D = \frac{1}{2} C \rho A v^2$$

where C is a constant.

Show that C has no unit.

[2]

- (b)** A raindrop falls vertically from rest. Assume that air resistance is negligible.

- (i)** On Fig. 1.1, sketch a graph to show the variation with time t of the velocity v of the raindrop for the first 1.0 s of the motion.

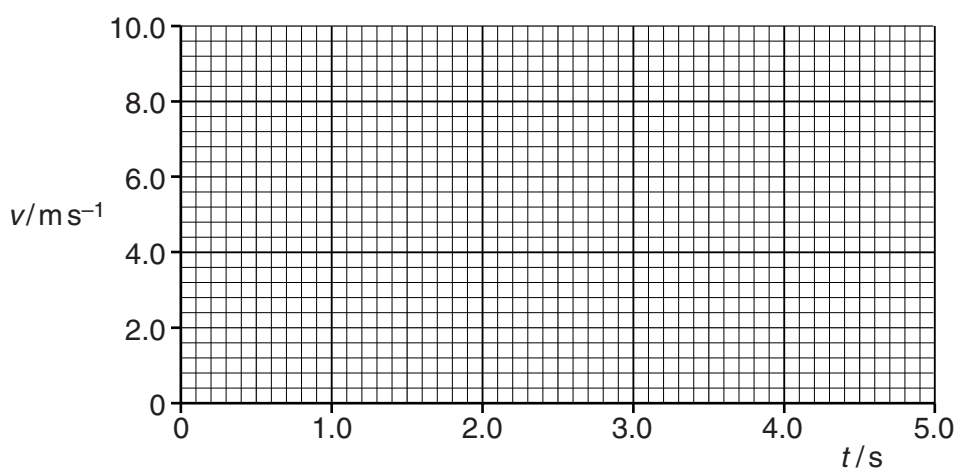


Fig. 1.1

[1]

- (ii)** Calculate the velocity of the raindrop after falling 1000 m.

velocity = ms^{-1} [2]

(c) In practice, air resistance on raindrops is not negligible because there is a drag force. This drag force is given by the expression in (a).

(i) State an equation relating the forces acting on the raindrop when it is falling at terminal velocity.

[1]

(ii) The raindrop has mass $1.4 \times 10^{-5} \text{ kg}$ and cross-sectional area $7.1 \times 10^{-6} \text{ m}^2$. The density of the air is 1.2 kg m^{-3} and the initial velocity of the raindrop is zero. The value of C is 0.60.

1. Show that the terminal velocity of the raindrop is about 7 m s^{-1} .

[2]

2. The raindrop reaches terminal velocity after falling approximately 10 m. On Fig. 1.1, sketch the variation with time t of velocity v for the raindrop. The sketch should include the first 5 s of the motion.

[2]

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- 17 Two planks of wood AB and BC are inclined at an angle of 15° to the horizontal. The two wooden planks are joined at point B, as shown in Fig. 2.1.

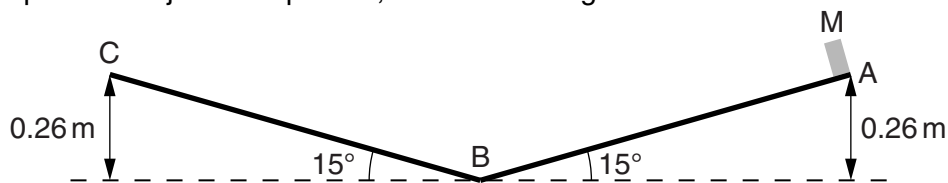


Fig. 2.1

A small block of metal M is released from rest at point A. It slides down the slope to B and up the opposite side to C. Points A and C are 0.26 m above B. Assume frictional forces are negligible.

- (a) (i) Describe and explain the acceleration of M as it travels from A to B and from B to C.

.....

 [3]

- (ii) Calculate the time taken for M to travel from A to B.

- (iii) Calculate the speed of M at B.

time = s [3]

speed = ms^{-1} [2]

- (b) The plank BC is adjusted so that the angle it makes with the horizontal is 30° . M is released from rest at point A and slides down the slope to B. It then slides a distance along the plank from B towards C.

Use the law of conservation of energy to calculate this distance. Explain your working.

distance = m [2]

- 18 (a)** A student walks from A to B along the path shown in Fig. 2.1.

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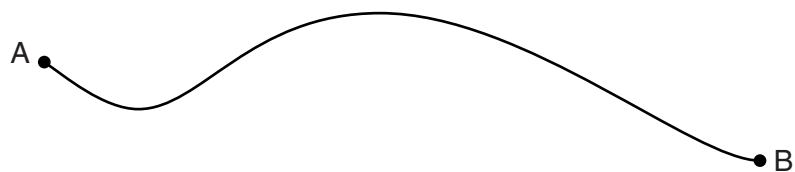


Fig. 2.1

The student takes time t to walk from A to B.

- (i) State the quantity, apart from t , that must be measured in order to determine the average value of

1. speed,

.....
..... [1]

2. velocity.

.....
..... [1]

- (ii) Define *acceleration*.

..... [1]

- (b) A girl falls vertically onto a trampoline, as shown in Fig. 2.2.

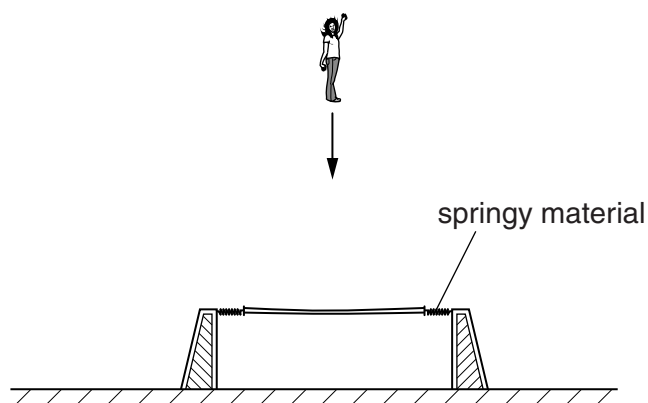


Fig. 2.2

The trampoline consists of a central section supported by springy material. At time $t = 0$ the girl starts to fall. The girl hits the trampoline and rebounds vertically. The variation with time t of velocity v of the girl is illustrated in Fig. 2.3.

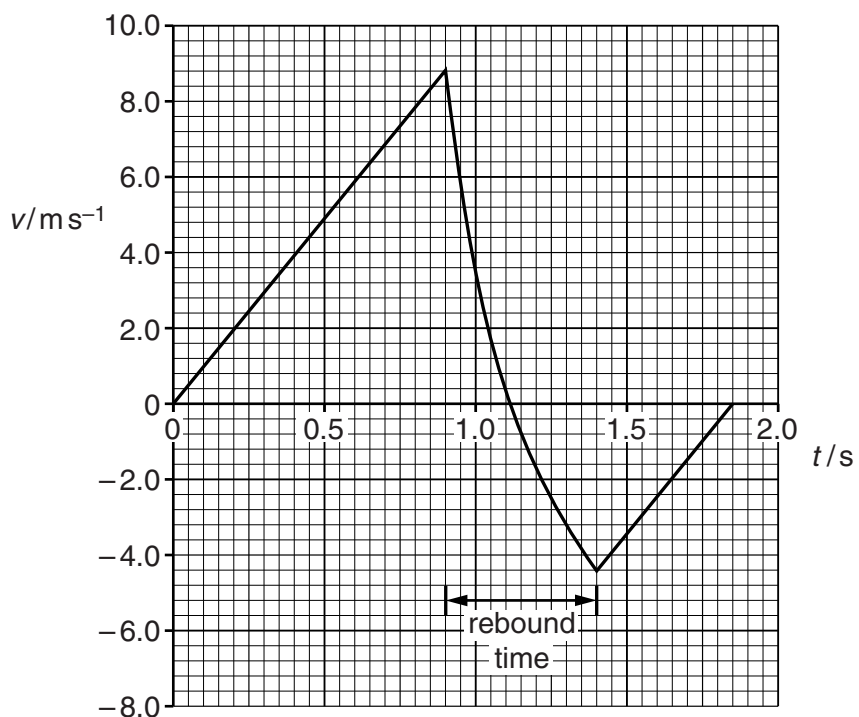


Fig. 2.3

For the motion of the girl, calculate

- (i) the distance fallen between time $t = 0$ and when she hits the trampoline,

distance = m [2]

- (ii) the average acceleration during the rebound.

acceleration = ms^{-2} [2]

- (c) (i) Use Fig. 2.3 to compare, without calculation, the accelerations of the girl before and after the rebound. Explain your answer.

.....
.....
.....[2]

- (ii) Use Fig. 2.3 to compare, without calculation, the potential energy of the girl at $t = 0$ and $t = 1.85\text{ s}$. Explain your answer.

.....
.....
.....[2]

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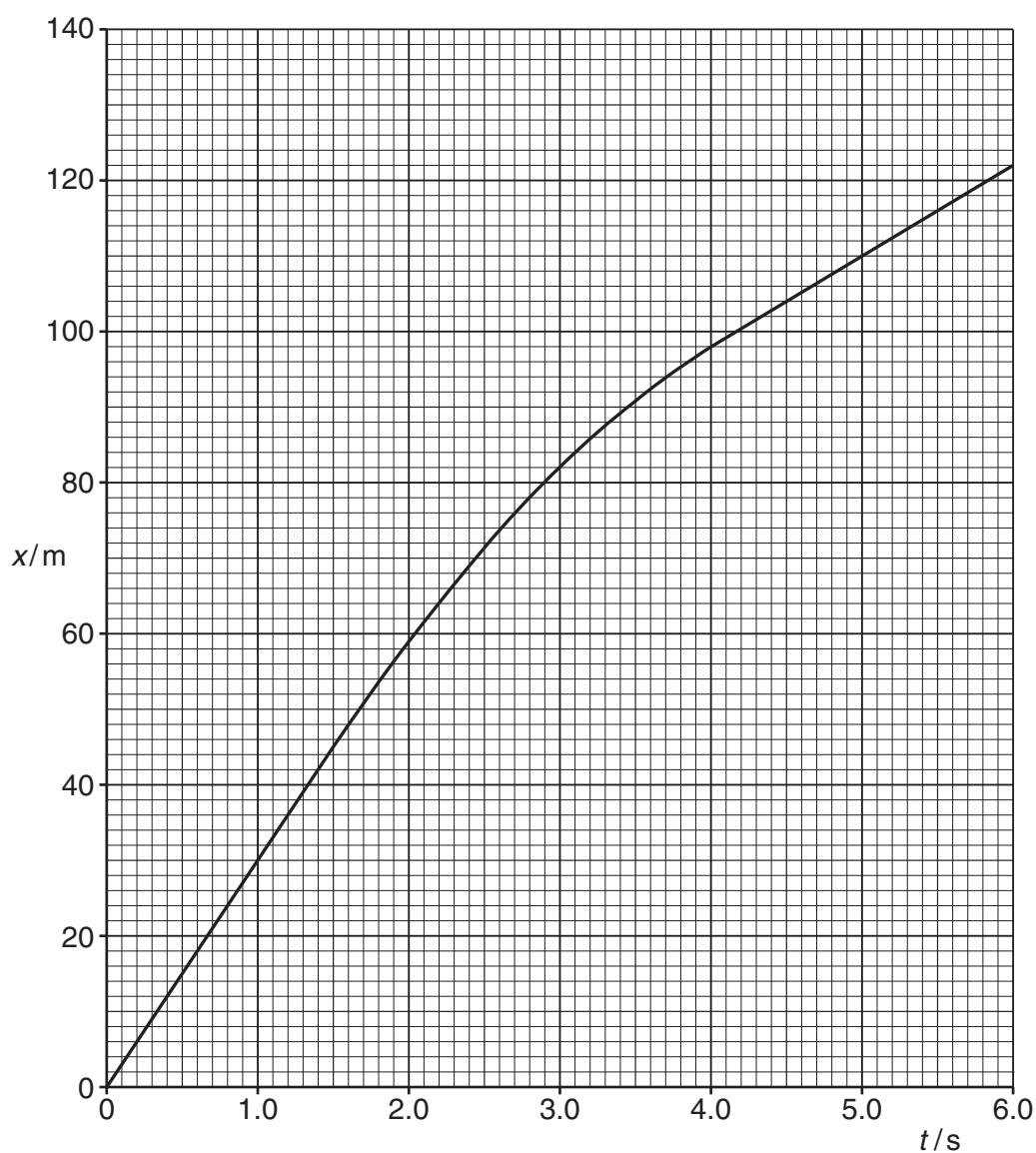
19 (a) Define(i) *velocity*,

.....
..... [1]

(ii) *acceleration*.

.....
..... [1]

(b) A car of mass 1500 kg travels along a straight horizontal road.
The variation with time t of the displacement x of the car is shown in Fig. 3.1.

**Fig. 3.1**

- (i) Use Fig. 3.1 to describe qualitatively the velocity of the car during the first six seconds of the motion shown.
Give reasons for your answers.

.....
.....
.....
.....
.....[3]

- (ii) Calculate the average velocity during the time interval $t = 0$ to $t = 1.5$ s.

average velocity = ms^{-1} [1]

- (iii) Show that the average acceleration between $t = 1.5$ s and $t = 4.0$ s is -7.2 ms^{-2} .

[2]

- (iv) Calculate the average force acting on the car between $t = 1.5$ s and $t = 4.0$ s.

force = N [2]

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20 (a) (i) Define *velocity*.

.....
[1]

(ii) Distinguish between *speed* and *velocity*.

.....
[2]

(b) A car of mass 1500 kg moves along a straight, horizontal road. The variation with time t of the velocity v for the car is shown in Fig. 1.1.

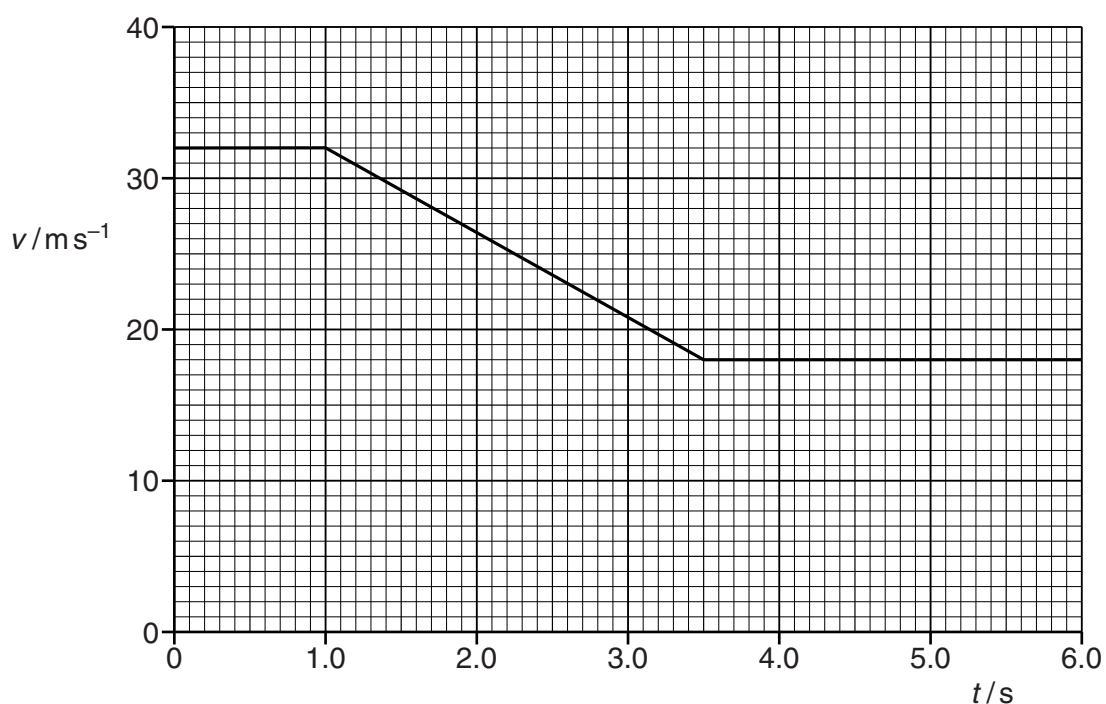


Fig. 1.1

The brakes of the car are applied from $t = 1.0$ s to $t = 3.5$ s.
 For the time when the brakes are applied,

(i) calculate the distance moved by the car,

distance = m [3]

- (ii) calculate the magnitude of the resultant force on the car.

resultant force = N [3]

- (c) The direction of motion of the car in (b) at time $t = 2.0$ s is shown in Fig. 1.2.

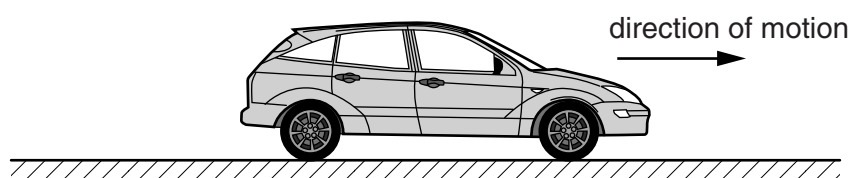


Fig. 1.2

On Fig. 1.2, show with arrows the directions of the acceleration (label this arrow A) and the resultant force (label this arrow F).

[1]

9702/23/M/J/14/Q2

- 21 (a)** Explain what is meant by a *scalar* quantity and by a *vector* quantity.

scalar:

.....

vector:

.....

[2]

- (b)** A ball leaves point P at the top of a cliff with a horizontal velocity of 15 ms^{-1} , as shown in Fig. 2.1.

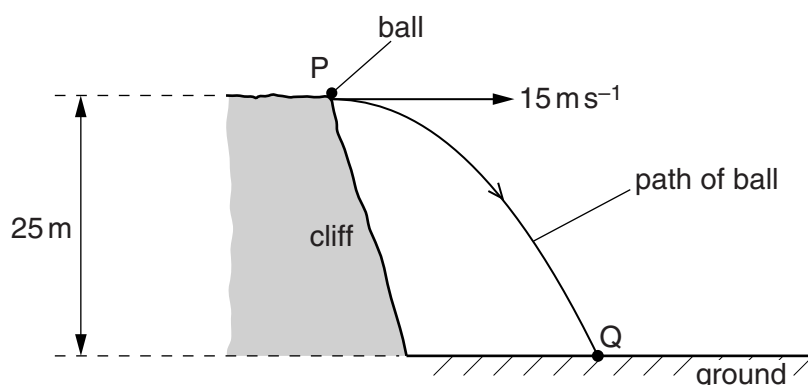


Fig. 2.1

The height of the cliff is 25 m. The ball hits the ground at point Q.
Air resistance is negligible.

- (i)** Calculate the vertical velocity of the ball just before it makes impact with the ground at Q.

vertical velocity = ms^{-1} [2]

- (ii)** Show that the time taken for the ball to fall to the ground is 2.3 s.

[1]

- (iii)** Calculate the magnitude of the displacement of the ball at point Q from point P.

displacement = m [4]

- (iv)** Explain why the distance travelled by the ball is different from the magnitude of the displacement of the ball.

.....

.....

.....[2]

9702/21/O/N/14/Q4

- 22** A trolley moves down a slope, as shown in Fig. 4.1.

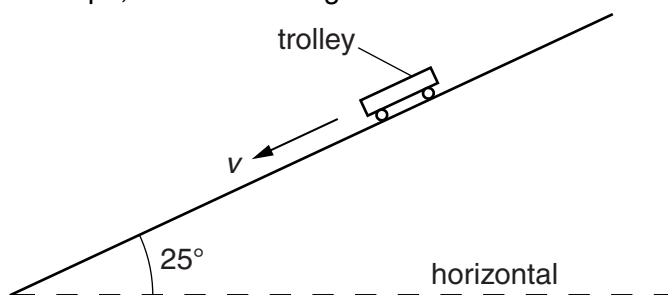


Fig. 4.1

The slope makes an angle of 25° with the horizontal. A constant resistive force F_R acts up the slope on the trolley.

At time $t = 0$, the trolley has velocity $v = 0.50 \text{ m s}^{-1}$ down the slope.

At time $t = 4.0 \text{ s}$, $v = 12 \text{ m s}^{-1}$ down the slope.

- (a) (i)** Show that the acceleration of the trolley down the slope is approximately 3 m s^{-2} .

[2]

- (ii)** Calculate the distance x moved by the trolley down the slope from time $t = 0$ to $t = 4.0 \text{ s}$

$x = \dots\dots\dots \text{ m}$ [2]

- (iii)** On Fig. 4.2, sketch the variation with time t of distance x moved by the trolley.

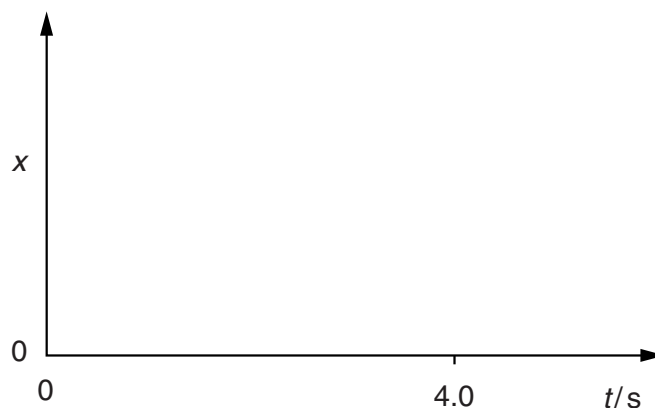


Fig. 4.2

[2]

- (b)** The mass of the trolley is 2.0 kg .

- (i)** Show that the component of the weight of the trolley down the slope is 8.3 N .

- (ii)** Calculate the resistive force F_R .

[1]

$F_R = \dots\dots\dots \text{ N}$ [2]

9702/22/O/N/14/Q2

- 23 A ball is thrown from A to B as shown in Fig. 2.1.

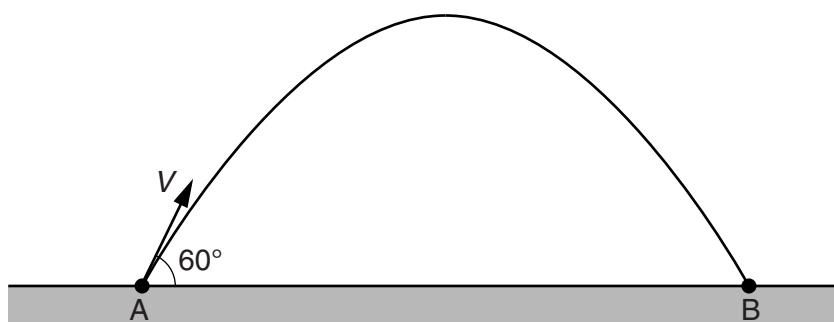


Fig. 2.1

The ball is thrown with an initial velocity V at 60° to the horizontal.

The variation with time t of the vertical component V_v of the velocity of the ball from $t = 0$ to $t = 0.60$ s is shown in Fig. 2.2.

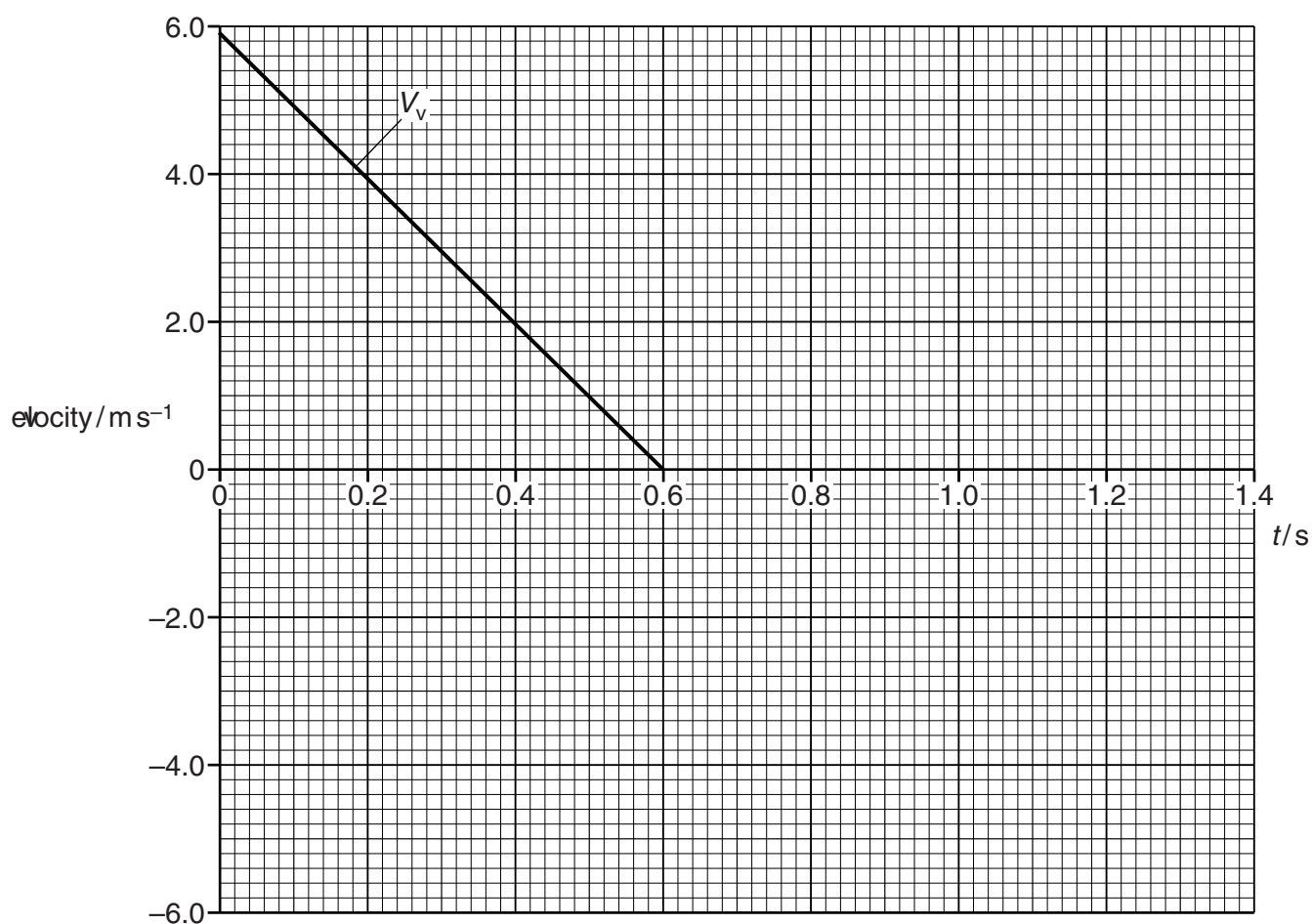


Fig. 2.2

Assume air resistance is negligible.

- (a) (i) Complete Fig. 2.2 for the time until the ball reaches B.

[2]

- (ii) Calculate the maximum height reached by the ball.

height =m [2]

- (iii) Calculate the horizontal component V_h of the velocity of the ball at time $t = 0$.

$V_h = \dots\dots\dots \text{ms}^{-1}$ [2]

- (iv) On Fig. 2.2, sketch the variation with t of V_h . **Label** this sketch V_h . [1]

- (b) The ball has mass 0.65 kg.

Calculate, for the ball,

- (i) the maximum kinetic energy,

maximum kinetic energy =J [3]

- (ii) the maximum potential energy above the ground.

maximum potential energy =J [2]

9702/21/M/J/15/Q2

- 24 (a)** Define *speed* and *velocity* and use these definitions to explain why one of these quantities is a scalar and the other is a vector.

speed:

velocity:

.....

.....[2]

- (b)** A ball is released from rest and falls vertically. The ball hits the ground and rebounds vertically, as shown in Fig. 2.1.

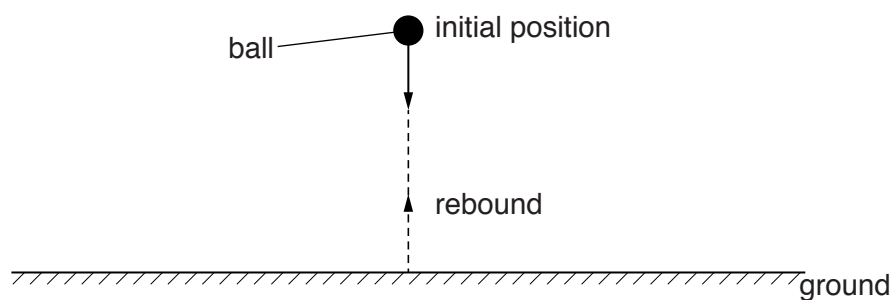


Fig. 2.1

The variation with time t of the velocity v of the ball is shown in Fig. 2.2.

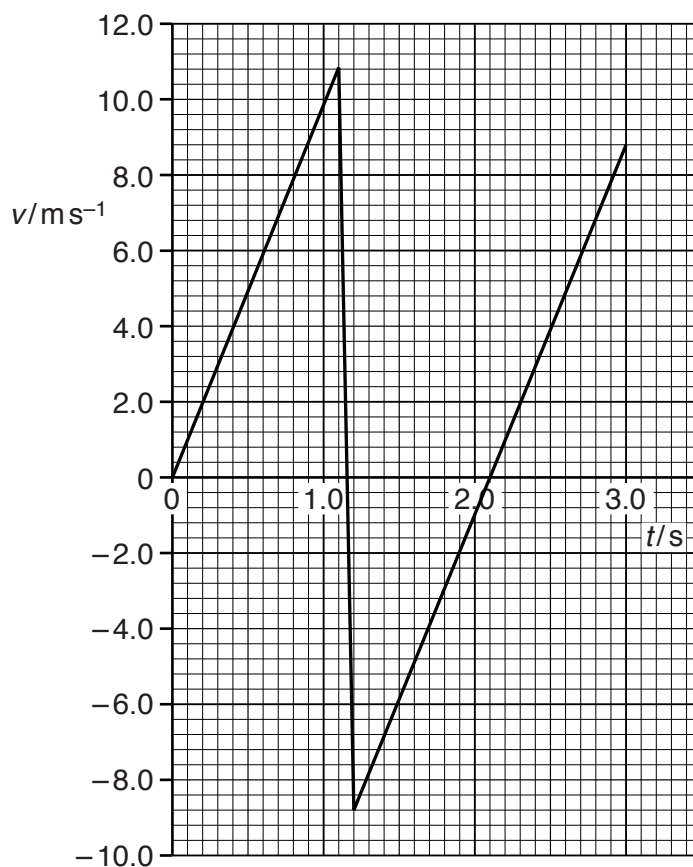


Fig. 2.2

Air resistance is negligible.

- (i) Without calculation, use Fig. 2.2 to describe the variation with time t of the velocity of the ball from $t = 0$ to $t = 2.1$ s.

.....

.....

.....

.....

.....[3]

- (ii) Calculate the acceleration of the ball after it rebounds from the ground. Show your working.

acceleration = ms^{-2} [3]

- (iii) Calculate, for the ball, from $t = 0$ to $t = 2.1$ s,

1. the distance moved,

distance = m [3]

2. the displacement from the initial position.

displacement = m [2]

(iv) On Fig. 2.3, sketch the variation with t of the speed of the ball.

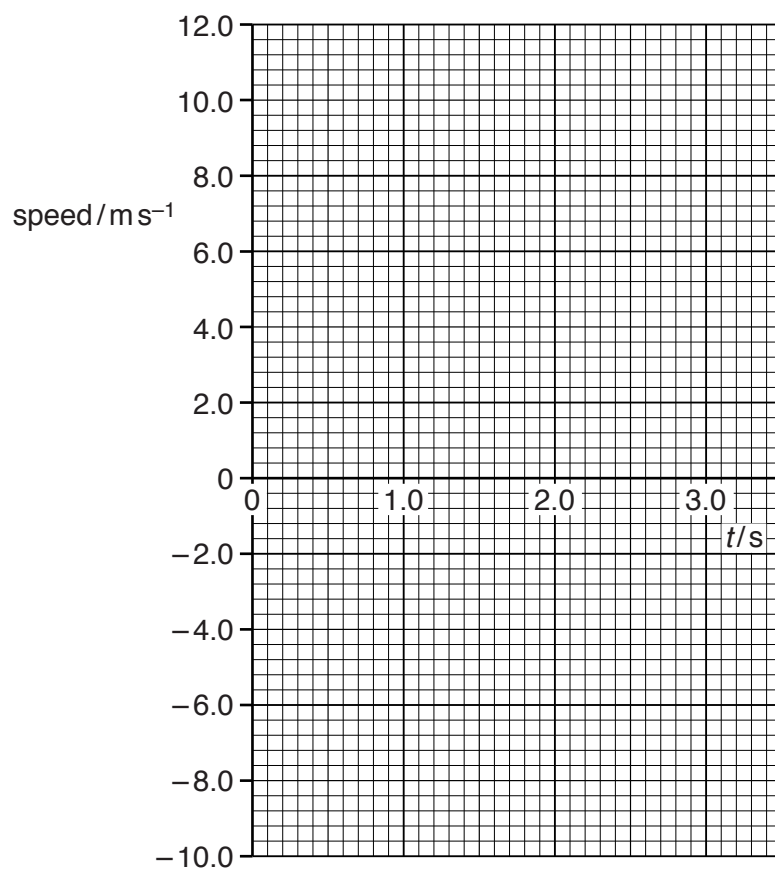


Fig. 2.3

[2]

9702/22/M/J/15/Q2

- 25** A stone is thrown vertically upwards. The variation with time t of the displacement s of the stone is shown in Fig. 2.1.

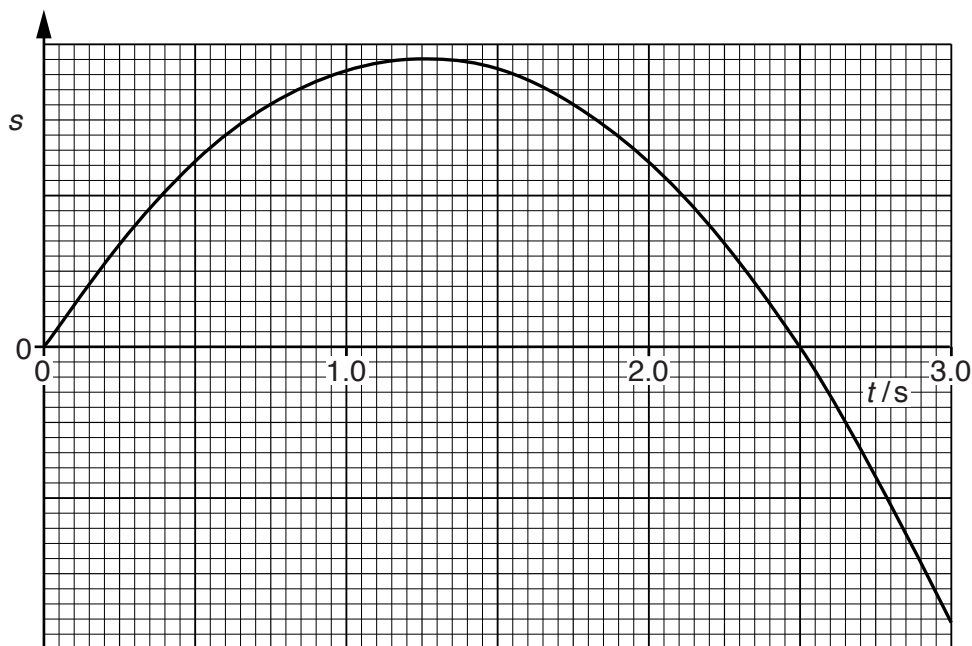


Fig. 2.1

- (a)** Use Fig. 2.1 to describe, without calculation, the speed of the stone from $t = 0$ to $t = 3.0$ s.

.....

.....

..... [2]

- (b)** Assume air resistance is negligible and therefore the stone has constant acceleration.

Calculate, for the stone,

- (i)** the speed at 3.0 s,

speed = ms^{-1}
[3]

(ii) the distance travelled from $t = 0$ to $t = 3.0$ s,

distance = m [3]

(iii) the displacement from $t = 0$ to $t = 3.0$ s.

displacement = m

direction [2]

(c) On Fig. 2.2, draw the variation with time t of the velocity v of the stone from $t = 0$ to $t = 3.0$ s.

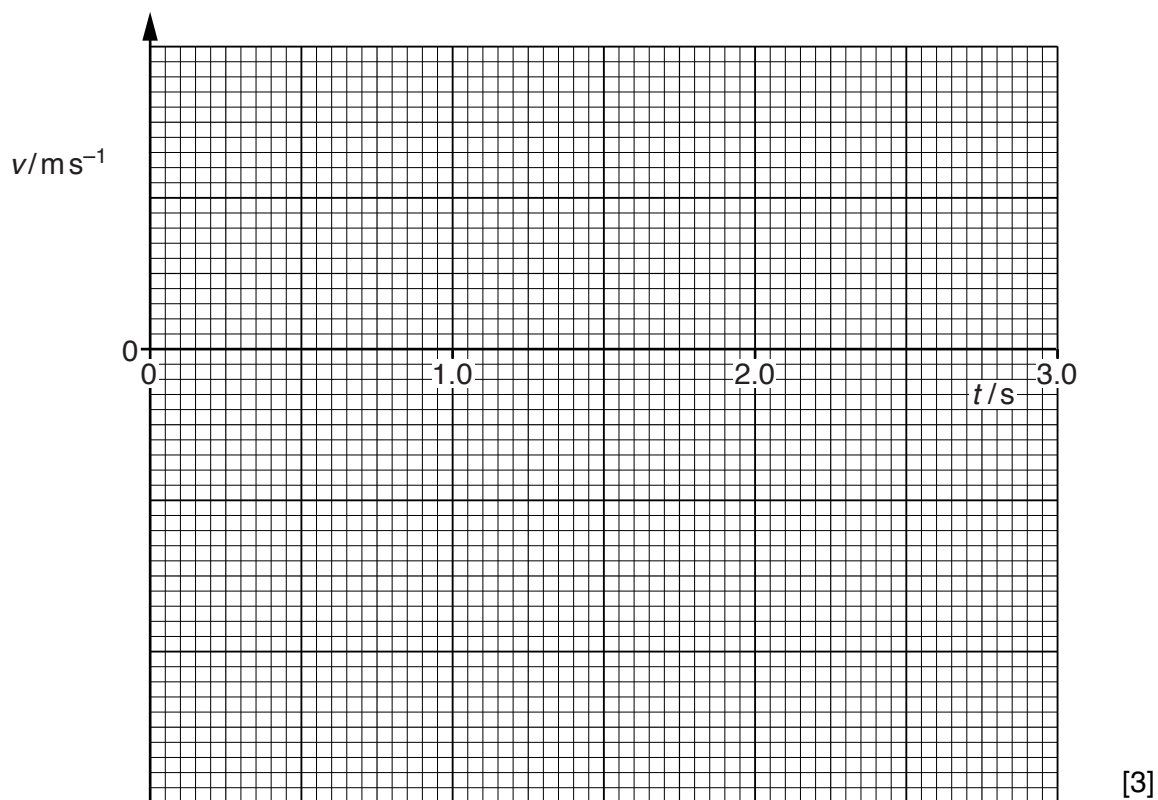


Fig. 2.2

9702/23/M/J/15/Q2

- 26** The variation with time t of the velocity v of a ball is shown in Fig. 2.1.

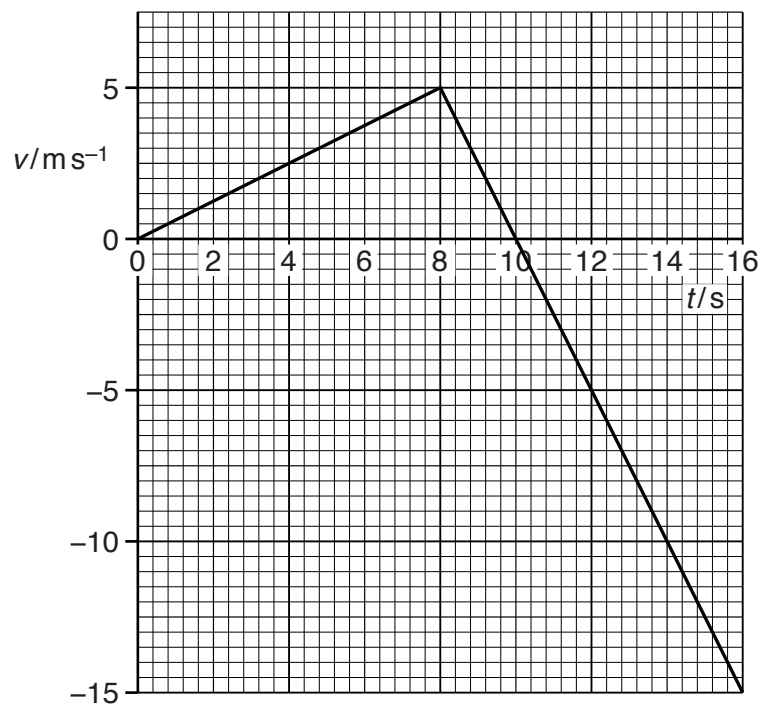


Fig. 2.1

The ball moves in a straight line from a point P at $t = 0$. The mass of the ball is 400 g.

- (a)** Use Fig. 2.1 to describe, without calculation, the velocity of the ball from $t = 0$ to $t = 16$ s.

.....

[2]

- (b)** Use Fig. 2.1 to calculate, for the ball,

- (i)** the displacement from P at $t = 10$ s,

displacement = m [2]

(ii) the acceleration at $t = 10\text{ s}$,

acceleration = ms^{-2} [2]

(iii) the maximum kinetic energy.

kinetic energy = J [2]

(c) Use your answers in (b)(i) and (b)(ii) to determine the time from $t = 0$ for the ball to return to P.

time = s [2]

9702/23/M/J/12/Q2

- 27** A motor drags a log of mass 452 kg up a slope by means of a cable, as shown in Fig. 2.1.

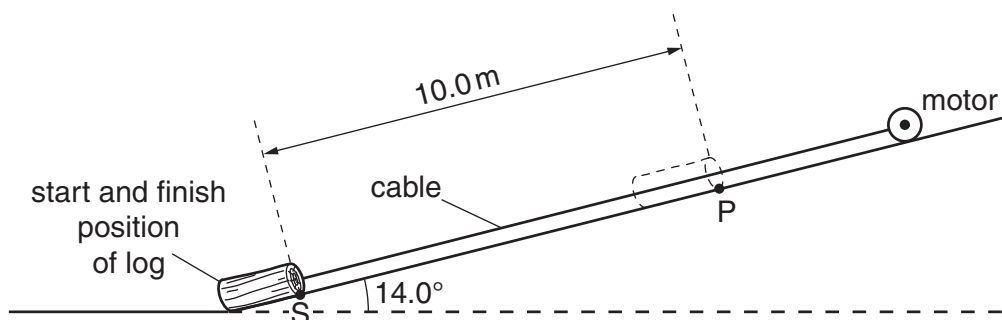


Fig. 2.1

The slope is inclined at 14.0° to the horizontal.

- (a) Show that the component of the weight of the log acting down the slope is 1070 N.

[1]

- (b) The log starts from rest. A constant frictional force of 525 N acts on the log. The log accelerates up the slope at 0.130 m s^{-2} .

- (i) Calculate the tension in the cable.

tension = N [3]

- (ii) The log is initially at rest at point S. It is pulled through a distance of 10.0m to point P.

Calculate, for the log,

1. the time taken to move from S to P,

time = s [2]

2. the magnitude of the velocity at P.

velocity = ms^{-1} [1]

- (c) The cable breaks when the log reaches point P. On Fig. 2.2, sketch the variation with time t of the velocity v of the log. The graph should show v from the start at S until the log returns to S. [4]



Fig. 2.2

28 (a) Define *acceleration*.

9702/22/F/M/16/Q2

.....
 [1]

(b) A ball is kicked from horizontal ground towards the top of a vertical wall, as shown in Fig. 2.1.

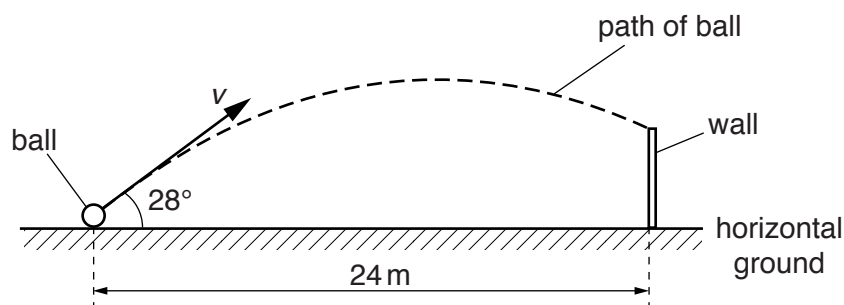


Fig. 2.1 (not to scale)

The horizontal distance between the initial position of the ball and the base of the wall is 24 m. The ball is kicked with an initial velocity v at an angle of 28° to the horizontal. The ball hits the top of the wall after a time of 1.5 s. Air resistance may be assumed to be negligible.

(i) Calculate the initial horizontal component v_x of the velocity of the ball.

$$v_x = \dots\dots\dots \text{ms}^{-1} \quad [1]$$

(ii) Show that the initial vertical component v_y of the velocity of the ball is 8.5 ms^{-1} .

[2]

(iii) Calculate the time taken for the ball to reach its maximum height above the ground.

$$\text{time} = \dots\dots\dots \text{s} \quad [2]$$

- (iv) The ball is kicked at time $t = 0$. On Fig. 2.2, sketch the variation with time t of the vertical component v_y of the velocity of the ball until it hits the wall. It may be assumed that velocity is positive when in the upwards direction.

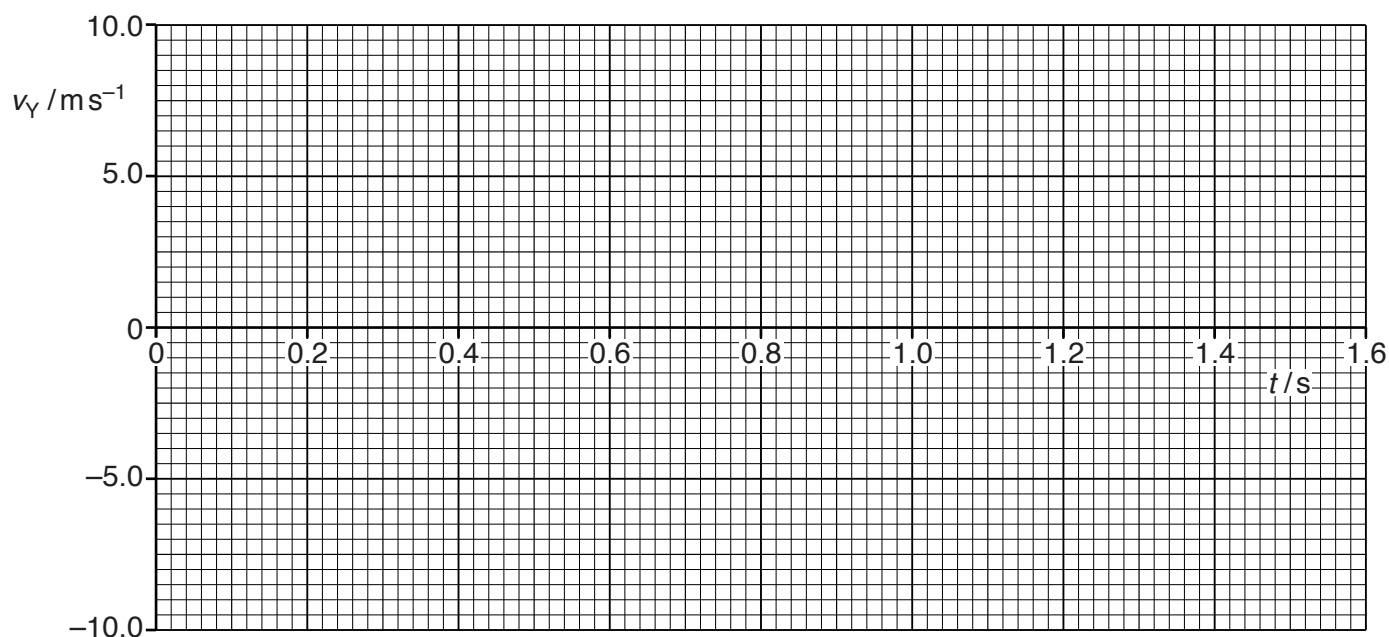


Fig. 2.2

[2]

- (c) (i) Use the information in (b) to determine the maximum height of the ball above the ground.

maximum height = m [2]

- (ii) The maximum gravitational potential energy of the ball above the ground is 22 J. Calculate the mass of the ball.

mass = kg [2]

- (d) A ball of greater mass is kicked with the same velocity as the ball in (b). State and explain the effect, if any, of the increased mass on the maximum height reached by the ball. Air resistance is still assumed to be negligible.

.....
 [1]

9702/21/M/J/16/Q2

- 29 A ball is thrown from a point P with an initial velocity u of 12 ms^{-1} at 50° to the horizontal, as illustrated in Fig. 2.1.

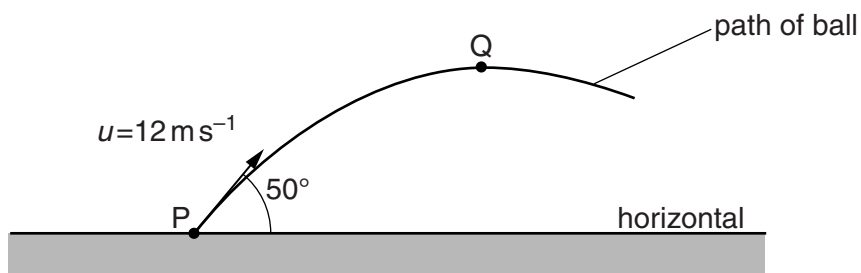


Fig. 2.1

The ball reaches maximum height at Q.

Air resistance is negligible.

(a) Calculate

- (i) the horizontal component of u ,

horizontal component = ms^{-1} [1]

- (ii) the vertical component of u .

vertical component = ms^{-1} [1]

- (b) Show that the maximum height reached by the ball is 4.3 m.

[2]

- (c) Determine the magnitude of the displacement PQ.

displacement = m [4]

30 (a) Define *velocity*.

9702/22/M/J/17/Q2

.....
 [1]

(b) A ball of mass 0.45 kg leaves the edge of a table with a horizontal velocity v , as shown in Fig. 2.1.

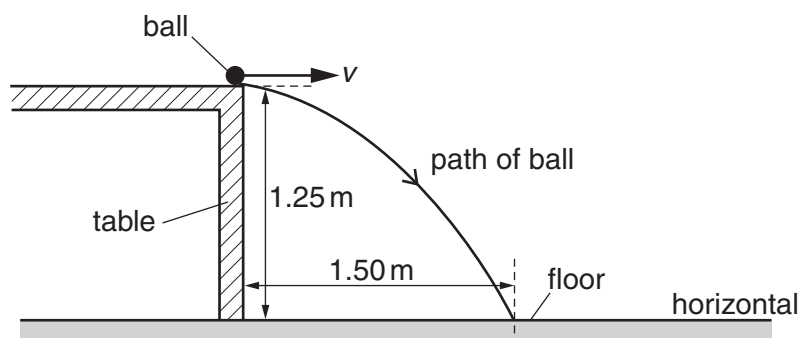


Fig. 2.1

The height of the table is 1.25 m. The ball travels a distance of 1.50 m horizontally before hitting the floor.

Air resistance is negligible.

Calculate, for the ball,

(i) the horizontal velocity v as it leaves the table,

$v = \dots\dots\dots \text{ms}^{-1}$ [3]

(ii) the velocity just as it hits the floor,

magnitude of velocity = $\dots\dots\dots \text{ms}^{-1}$

angle to the horizontal = $\dots\dots\dots^\circ$
 [4]

9702/21/O/N/17/Q2

- 31 The variation with time t of the velocity v of two cars P and Q is shown in Fig. 2.1.

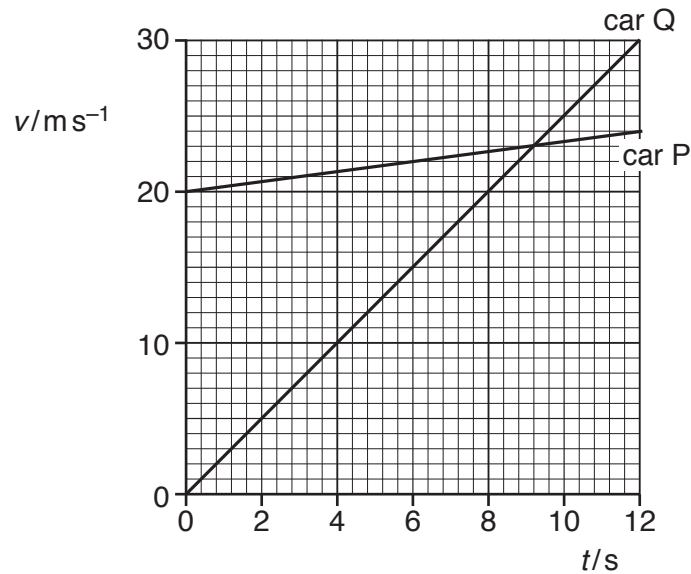


Fig. 2.1

The cars travel in the same direction along a straight road.
Car P passes car Q at time $t = 0$.

- (a) The speed limit for cars on the road is 100 km h^{-1} . State and explain whether car Q exceeds the speed limit.

.....[1]

- (b) Calculate the acceleration of car P.

acceleration = ms^{-2} [2]

- (c) Determine the distance between the two cars at time $t = 12\text{ s}$.

distance = m [3]

- (d) From time $t = 12\text{ s}$, the velocity of each car remains constant at its value at $t = 12\text{ s}$.

Determine the time t at which car Q passes car P.

$t =$ s [2]

9702/22/M/J/18/Q3

- 32** A child on a sledge slides down a steep hill and then travels in a straight line up an ice-covered slope, as illustrated in Fig. 3.1.

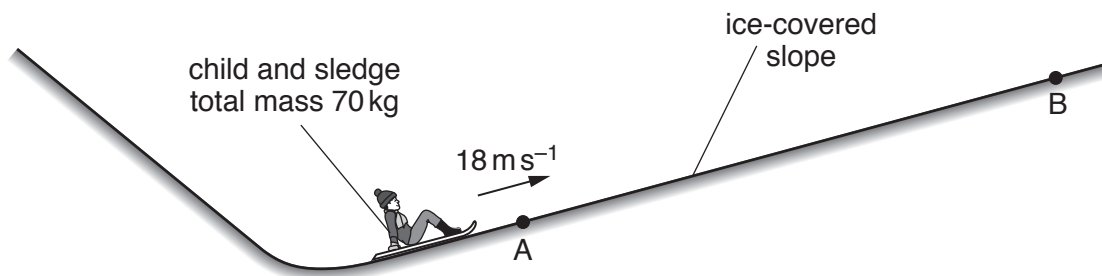


Fig. 3.1 (not to scale)

The sledge passes point A with speed 18 m s^{-1} at time $t = 0$ and then comes to rest at point B. The child applies a brake to the sledge at point B. The brake does not keep the sledge stationary and it immediately slides back down the slope towards A.

The variation with time t of the velocity v of the sledge from $t = 0$ to $t = 24 \text{ s}$ is shown in Fig. 3.2.

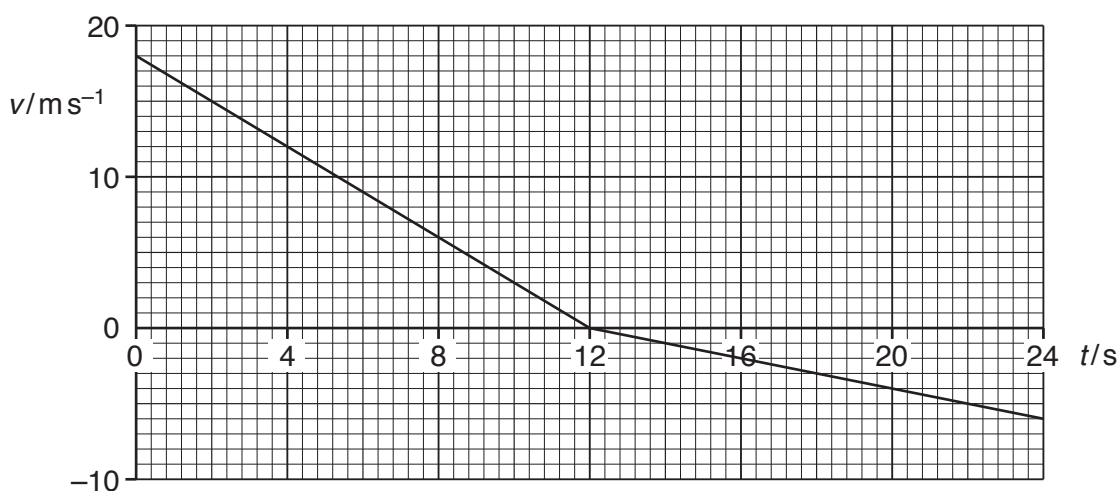


Fig. 3.2

- (a)** State the time taken for the sledge to travel from A to B.

time = s [1]

- (b)** Determine the displacement of the sledge up the slope from point A at time $t = 24$ s.

displacement =m [3]

- (c)** Show that the acceleration of the sledge as it moves from B back towards A is 0.50 m s^{-2} .

[2]

- (d)** The child and sledge have a total mass of 70 kg. The component of the total weight of the child and sledge that acts down the slope is 80 N.

Determine

- (i)** the frictional force on the sledge as it moves from B towards A,

frictional force = N [2]

- (ii)** the angle θ of the slope to the horizontal.

$\theta =$ $^{\circ}$ [2]

- (e) The child on the sledge blows a whistle between $t = 4.0\text{ s}$ and $t = 8.0\text{ s}$. The whistle emits sound of frequency 900 Hz . The speed of the sound in the air is 340 m s^{-1} . A man standing at point A hears the sound.

Use Fig. 3.2 to

- (i) determine the initial frequency of the sound heard by the man,

initial frequency = Hz [2]

- (ii) describe and explain qualitatively the variation, if any, in the frequency of the sound heard by the man.

.....
[1]

- 33 Two vertical metal plates in a vacuum are separated by a distance of 0.12 m . Fig. 4.1 shows a side view of this arrangement.

9702/23/M/J/19/Q4

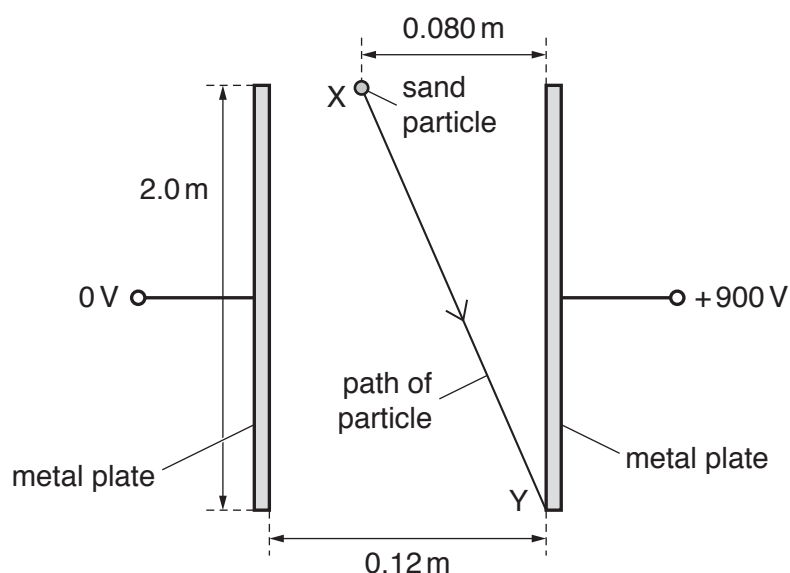


Fig. 4.1 (not to scale)

Each plate has a length of 2.0 m . The potential difference between the plates is 900 V . The electric field between the plates is uniform.

A negatively charged sand particle is released from rest at point X, which is a horizontal distance of 0.080 m from the top of the positively charged plate. The particle then travels in a straight line and collides with the positively charged plate at its lowest point Y, as illustrated in Fig. 4.1.

- (a) Describe the pattern of the field lines (lines of force) between the plates.

.....

[2]

- (b) State the names of the **two** forces acting on the particle as it moves from X to Y.

.....[1]

- (c) By considering the vertical motion of the sand particle, show that the time taken for the particle to move from X to Y is 0.64 s.

[2]

- (d) Calculate the horizontal component of the acceleration of the particle.

horizontal component of acceleration = ms^{-2} [2]

- (e) (i)** Calculate the magnitude of the electric field strength.

electric field strength = NC^{-1} [2]

- (ii)** The sand particle has mass m and charge q . Use your answers in **(d)** and **(e)(i)** to determine the ratio $\frac{q}{m}$.

ratio = C kg^{-1} [2]

- (f)** Another particle has a smaller magnitude of the ratio $\frac{q}{m}$ than the sand particle. This particle is also released from point X.

For the movement of this particle, state the effect, if any, of the decreased magnitude of the ratio on:

- (i)** the vertical component of the acceleration

.....[1]

- (ii)** the horizontal component of the acceleration.

.....[1]

9702/21/O/N/19/Q2

- 34** A small charged glass bead of weight $5.4 \times 10^{-5} \text{ N}$ is initially at rest at point A in a vacuum. The bead then falls through a uniform horizontal electric field as it moves in a straight line to point B, as illustrated in Fig. 2.1.

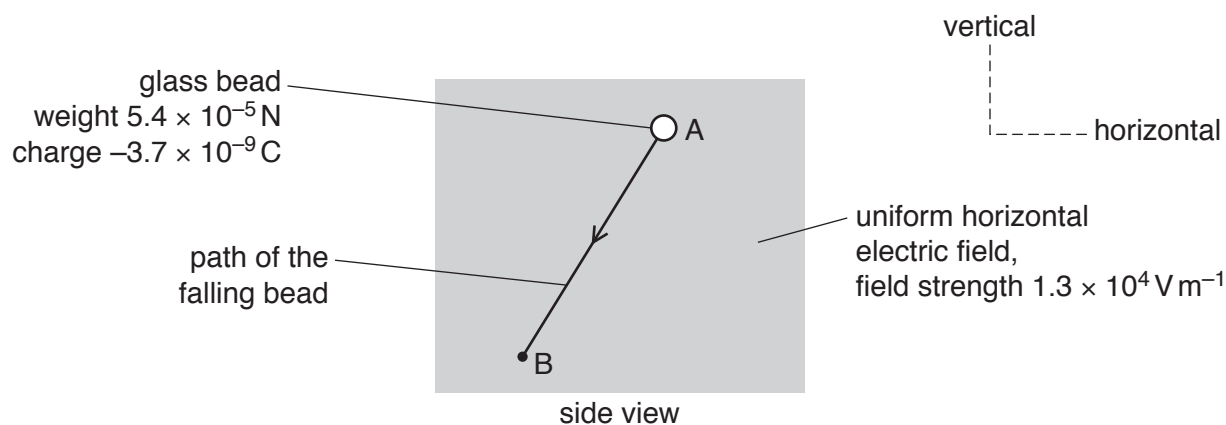


Fig. 2.1 (not to scale)

The electric field strength is $1.3 \times 10^4 \text{ V m}^{-1}$. The charge on the bead is $-3.7 \times 10^{-9} \text{ C}$.

- (a)** Describe how two metal plates could be used to produce the electric field. Numerical values are not required.

.....

.....

..... [2]

- (b)** Determine the magnitude of the electric force acting on the bead.

electric force = N [2]

- (c) Use your answer in (b) and the weight of the bead to show that the resultant force acting on it is $7.2 \times 10^{-5} \text{ N}$.

[1]

- (d) Explain why the resultant force on the bead of $7.2 \times 10^{-5} \text{ N}$ is constant as the bead moves along path AB.

.....

.....

.....

..... [2]

- (e) (i) Calculate the magnitude of the acceleration of the bead along the path AB.

acceleration = ms^{-2} [2]

- (ii) The path AB has length 0.58 m.

Use your answer in (i) to determine the speed of the bead at point B.

speed = ms^{-1} [2]

9702/22/O/N/19/Q2

35 (a) Define *acceleration*.

..... [1]

(b) A steel ball of diameter 0.080 m is released from rest and falls vertically in air, as illustrated in Fig. 2.1.

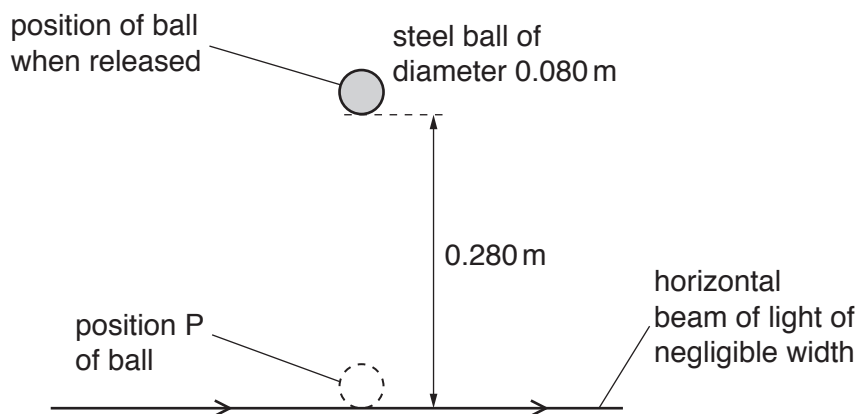


Fig. 2.1 (not to scale)

A horizontal beam of light of negligible width is a vertical distance of 0.280 m below the bottom of the ball when it is released. The ball falls through and breaks the beam of light.

(i) Explain why the force due to air resistance acting on the ball may be neglected when calculating the time taken for the ball to reach the beam of light.

.....
 [1]

(ii) Calculate the time taken for the ball to fall from rest to position P where the bottom of the ball touches the beam of light.

time taken = s [2]

- (iii) Determine the time interval during which the beam of light is broken by the ball.

time interval = s [2]

- (c) A different ball is released from the same position as the steel ball in (b). This ball has the same diameter but a much lower density. For this ball, the force due to air resistance cannot be neglected as the ball falls.

State and explain the change, if any, to the time interval during which the beam of light is broken by the ball.

.....
.....
.....
..... [2]

9702/23/O/N/19/Q2

- 36 (a) State what is meant by *work done*.

.....
 [1]

- (b) A lift (elevator) of weight 13.0 kN is connected by a cable to a motor, as shown in Fig. 2.1.

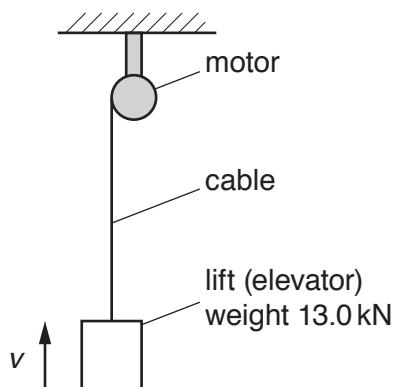


Fig. 2.1

The lift is pulled up a vertical shaft by the cable. A constant frictional force of 2.0 kN acts on the lift when it is moving. The variation with time t of the speed v of the lift is shown in Fig. 2.2.

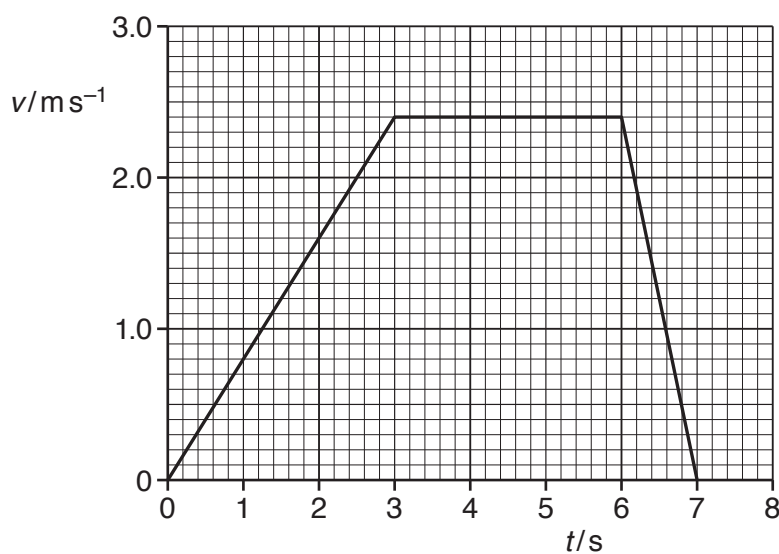


Fig. 2.2

(i) Use Fig. 2.2 to determine:

1. the acceleration of the lift between time $t = 0$ and $t = 3.0$ s

acceleration = m s^{-2} [2]

2. the work done by the motor to raise the lift between time $t = 3.0$ s and $t = 6.0$ s.

work done = J [2]

(ii) The motor has an efficiency of 67%. The tension in the cable is 1.6×10^4 N at time $t = 2.5$ s.

Determine the input power to the motor at this time.

input power = W [3]

(iii) State and explain whether the increase in gravitational potential energy of the lift from time $t = 0$ to $t = 7.0$ s is less than, the same as, or greater than the work done by the motor. A calculation is not required.

.....
 [1]

9702/22/M/J/20/Q1

- 37 (a) Define *velocity*.

.....
 [1]

- (b) The drag force F_D acting on a car moving with speed v along a straight horizontal road is given by

$$F_D = v^2 Ak$$

where k is a constant and A is the cross-sectional area of the car.

Determine the SI base units of k .

SI base units [2]

- (c) The value of k , in SI base units, for the car in (b) is 0.24. The cross-sectional area A of the car is 5.1 m^2 .

The car is travelling with a constant speed along a straight road and the output power of the engine is $4.8 \times 10^4 \text{ W}$. Assume that the output power of the engine is equal to the rate at which the drag force F_D is doing work against the car.

Determine the speed of the car.

speed = ms^{-1} [3]

9702/23/M/J/20/Q1

- 38 (a) State **one** similarity and **one** difference between *distance* and *displacement*.

similarity:

.....

difference:

.....

[2]

- (b) A student takes several measurements of the same quantity. This set of measurements has high precision, but low accuracy.

Describe what is meant by:

- (i) *high precision*

.....

..... [1]

- (ii) *low accuracy*.

.....

..... [1]

39. 9702/21/O/N/20/Q2

A small block is lifted vertically upwards by a toy aircraft, as illustrated in Fig. 2.1.

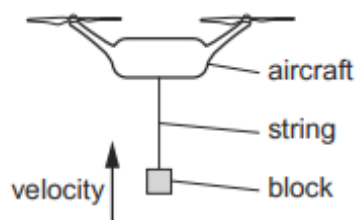


Fig. 2.1

As the block is moving upwards, the string breaks at time $t = 0$. The block initially continues moving upwards and then falls and hits the ground at time $t = 0.90\text{ s}$. The variation with time t of the velocity v of the block is shown in Fig. 2.2.

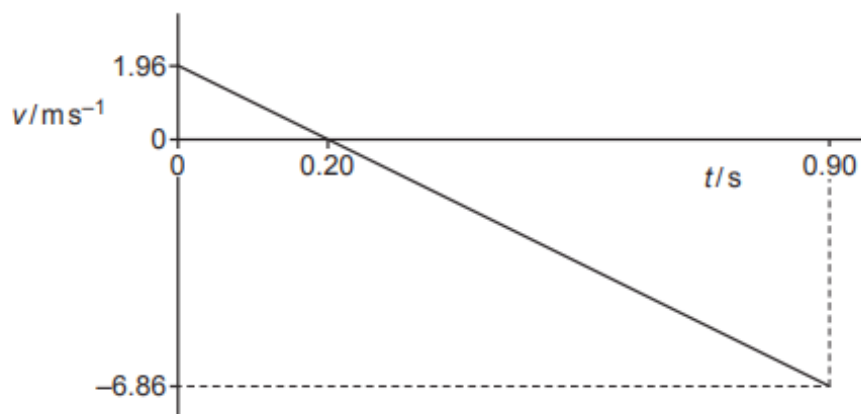


Fig. 2.2

Air resistance is negligible.

- (a) State the feature of the graph in Fig. 2.2 that shows the block has a constant acceleration.

..... [1]

- (b) Use Fig. 2.2 to determine the height of the block above the ground when the string breaks at time $t = 0$.

height = m [3]

- (c) The block has a weight of 0.86 N.

Calculate the difference in gravitational potential energy of the block between time $t = 0$ and time $t = 0.90$ s.

difference in gravitational potential energy = J [2]

- (d) On Fig. 2.3, sketch a line to show the variation of the distance moved by the block with time t from $t = 0$ to $t = 0.20$ s. Numerical values of distance are not required.

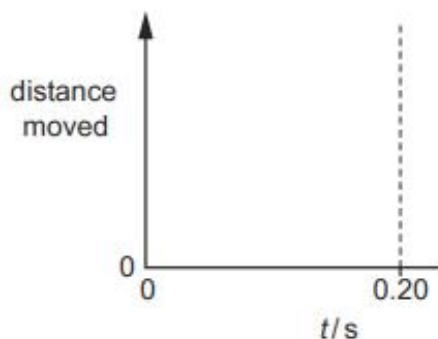


Fig. 2.3

[2]

- (e) A block of greater mass is now released from the same height with the same upward velocity. Air resistance is still negligible.

State and explain the effect, if any, of the increased mass on the speed with which the block hits the ground.

.....

..... [1]

[Total: 9]

40. 9702/22/O/N/20/Q1

- (a) Complete Table 1.1 by putting a tick (✓) in the appropriate column to indicate whether the listed quantities are scalars or vectors.

Table 1.1

quantity	scalar	vector
acceleration		
density		
temperature		
momentum		

[2]

- (b) A toy train moves along a straight section of track. Fig. 1.1 shows the variation with time t of the distance d moved by the train.

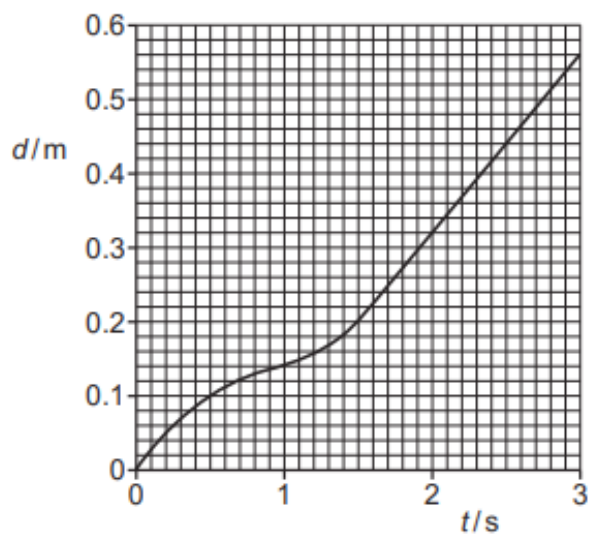


Fig. 1.1

- (i) Describe qualitatively the motion of the train between time $t = 0$ and time $t = 1.0$ s.

.....
 [1]

(ii) Determine the speed of the train at time $t = 2.0$ s.

speed = ms^{-1} [2]

(c) The straight section of track in (b) is part of the loop of track shown in Fig. 1.2.



Fig. 1.2

The train completes exactly one lap of the loop.

State and explain the average velocity of the train over the one complete lap.

.....
.....
..... [1]

[Total: 6]

41. 9702/23/O/N/20/Q3

A ball is fired horizontally with a speed of 41.0 m s^{-1} from a stationary cannon at the top of a hill. The ball lands on horizontal ground that is a vertical distance of 57 m below the cannon, as shown in Fig. 3.1.

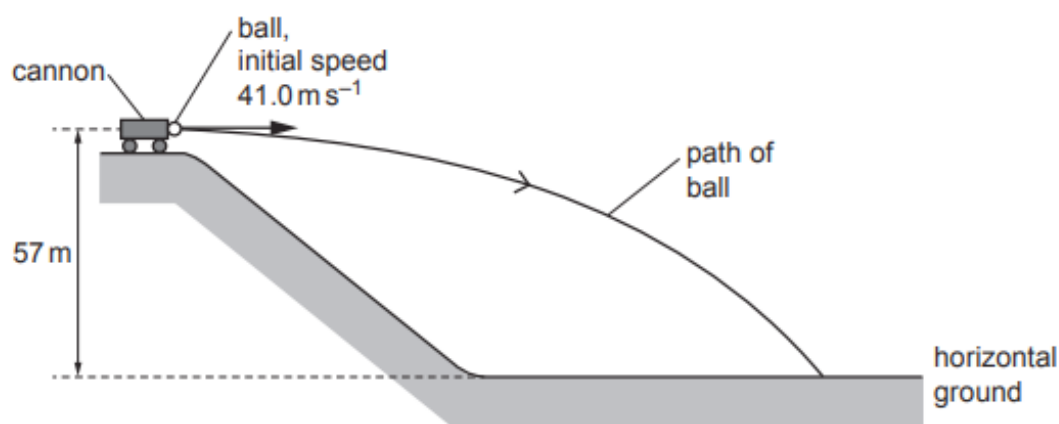


Fig. 3.1 (not to scale)

Assume air resistance is negligible.

- (a) Show that the time taken for the ball to reach the ground, after being fired, is 3.4 s.

[2]

- (b) Calculate the horizontal distance of the ball from the cannon at the point where the ball lands on the ground.

horizontal distance = m [1]

- (c) Determine the magnitude of the displacement of the ball from the cannon at the point where the ball lands on the ground.

displacement = m [2]

- (d) The ball leaves the cannon at time $t = 0$.

On Fig. 3.2, sketch a graph to show the variation of the magnitude v of the vertical component of the velocity of the ball with time t from $t = 0$ to $t = 3.4$ s. Numerical values are not required.

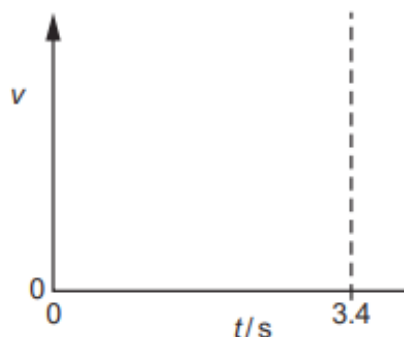


Fig. 3.2

[1]

- (e) The cannon recoils horizontally with a speed of 0.340 ms^{-1} when it fires the ball. The total mass of the ball and the cannon is 1480 kg . Assume that no external horizontal forces act on the ball-cannon system.

Determine, to three significant figures, the mass of the ball.

mass = kg [2]

- (f) The cannon now fires a ball of smaller mass. Assume that air resistance is still negligible.

State and explain the change, if any, to the graph in Fig. 3.2 due to the decreased mass of the ball.

.....

 [2]

[Total: 10]

42. 9702/22/F/M/21/Q1

(a) Complete Table 1.1 by stating whether each of the quantities is a vector or a scalar.

Table 1.1

quantity	vector or scalar
acceleration	
power	
work	

[2]

(b) The variation with time t of the velocity v of an object is shown in Fig. 1.1.

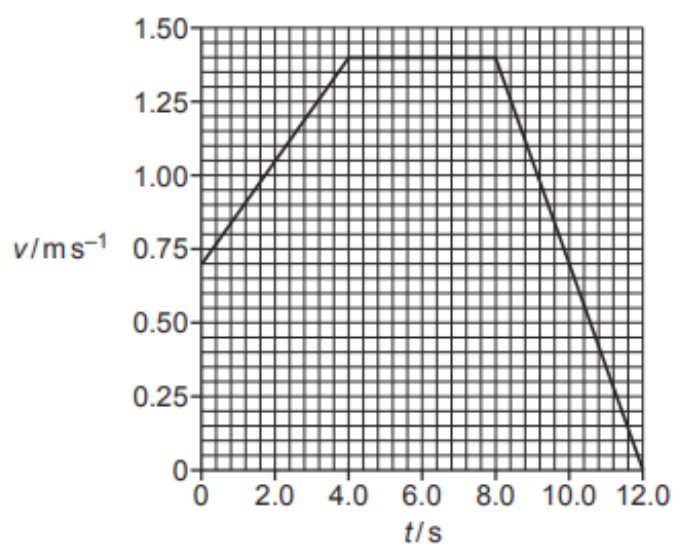


Fig. 1.1

(i) Determine the acceleration of the object from time $t = 0$ to time $t = 4.0$ s.

acceleration = ms^{-2} [2]

- (ii) Determine the distance moved by the object from time $t = 0$ to time $t = 4.0$ s.

distance = m [2]

- (c) (i) Define *force*.

.....
 [1]

- (ii) The motion represented in Fig. 1.1 is caused by a resultant force F acting on the object.

On Fig. 1.2, sketch the variation of F with time t from $t = 0$ to $t = 12.0$ s.
 Numerical values of F are not required.

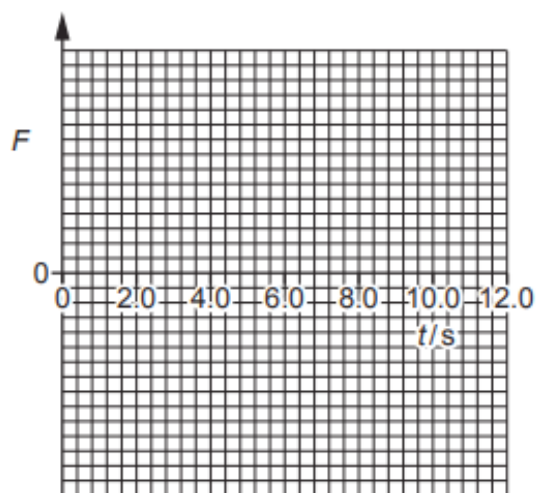


Fig. 1.2

[3]

[Total: 10]

43. 9702/23/M/J/21/Q2

(a) Define *acceleration*.

.....
..... [1]

(b) A stone falls vertically from the top of a cliff. Fig. 2.1 shows the variation with time t of the velocity v of the stone.

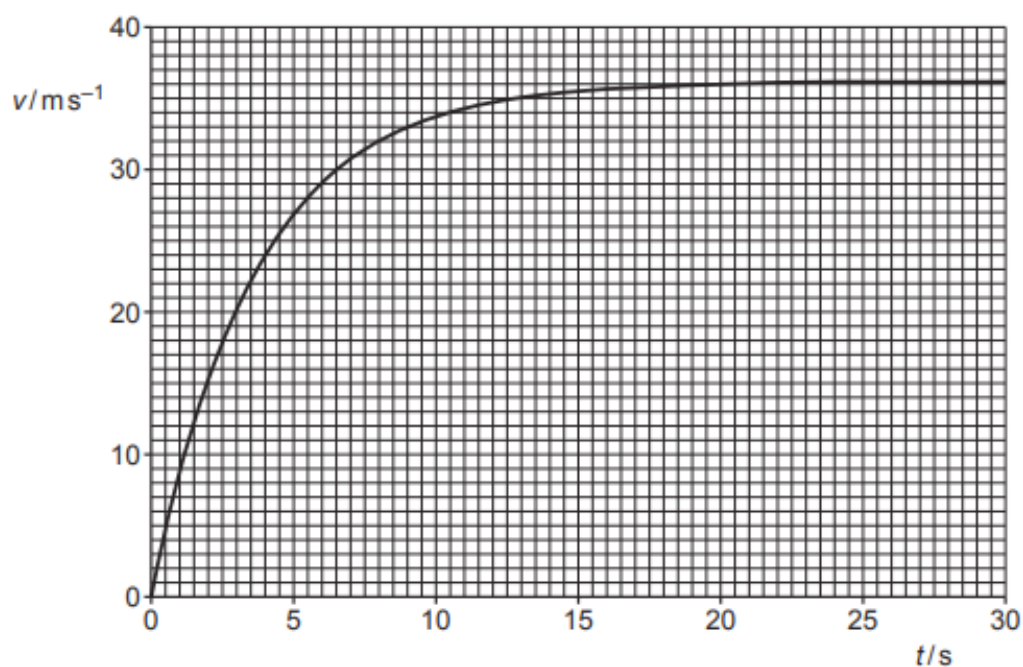


Fig. 2.1

(i) Explain, with reference to forces acting on the stone, the shape of the curve in Fig. 2.1.

.....
.....
.....
.....
..... [3]

(ii) Use Fig. 2.1 to determine the speed of the stone when the resultant force on it is zero.

speed = ms^{-1} [1]

- (iii) Use Fig. 2.1 to calculate the approximate height through which the stone falls between $t = 0$ and $t = 30$ s.

height = m [3]

- (iv) On Fig. 2.2, sketch the variation with t of the acceleration a of the stone between $t = 0$ and $t = 30$ s.

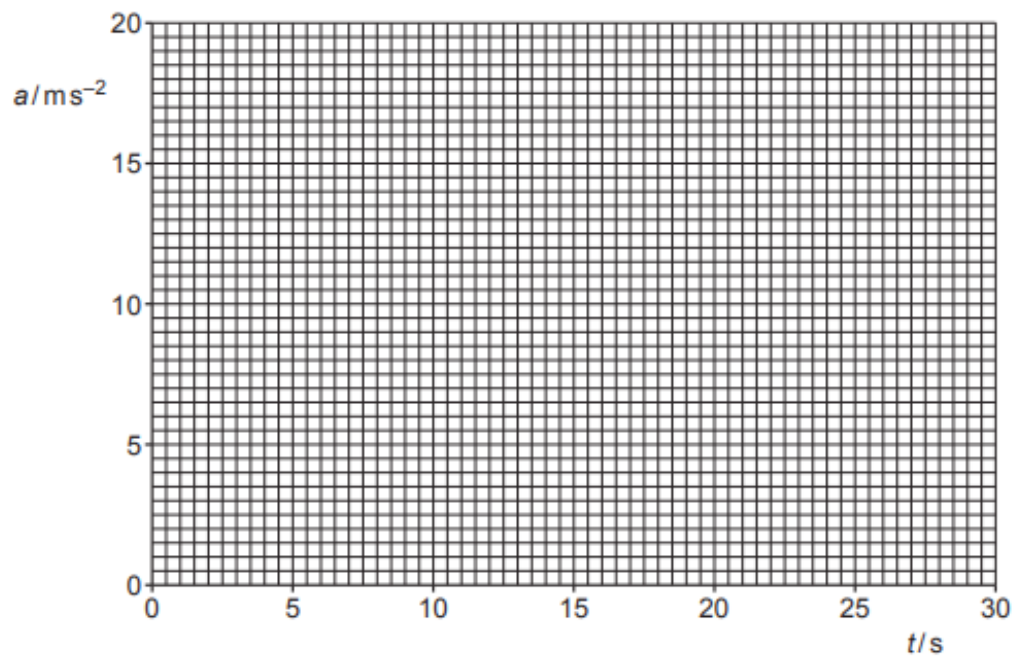


Fig. 2.2

[3]

[Total: 11]

44. 9702/22/O/N/21/Q4

A child sits on the ground next to a remote-controlled toy car. At time $t = 0$, the car begins to move in a straight line directly away from the child. The variation with time t of the velocity of the car along this line is shown in Fig. 4.1.

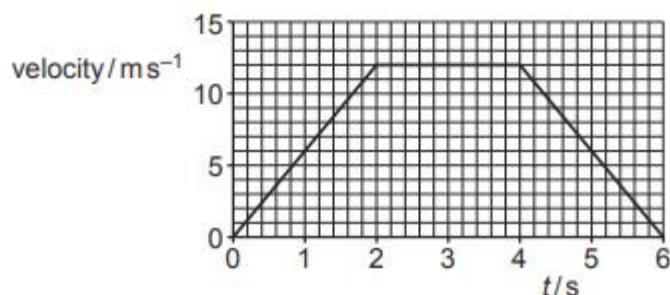


Fig. 4.1

The car's horn continually emits sound of frequency 925 Hz between time $t = 0$ and time $t = 6.0$ s. The speed of the sound in the air is 338 m s^{-1} .

- (a) Describe qualitatively the variation, if any, in the frequency of the sound heard, by the child, that was emitted from the car horn:

- (i) from time $t = 0$ to time $t = 2.0$ s

..... [1]

- (ii) from time $t = 4.0$ s to time $t = 6.0$ s.

..... [1]

- (b) Determine the frequency, to three significant figures, of the sound heard, by the child, that was emitted from the car horn at time $t = 3.0$ s.

frequency = Hz [2]

(c) Determine the time taken for the sound emitted at time $t = 4.0$ s to travel to the child.

time taken = s [2]

[Total: 6]

45. 9702/23/O/N/21/Q3

(a) Define *velocity*.

.....
 [1]

(b) A remote-controlled toy aircraft is flying horizontally in a wind. Fig. 3.1 shows the velocity vectors, to scale, of the wind and of the aircraft in still air.

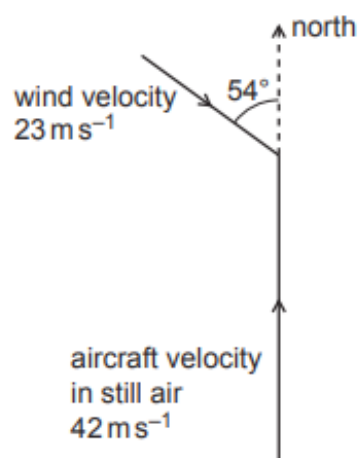


Fig. 3.1

The velocity of the aircraft in still air is 42 m s^{-1} to the north. The velocity of the wind is 23 m s^{-1} in a direction of 54° east of south.

Determine the magnitude of the resultant velocity of the aircraft.

magnitude of velocity = m s^{-1} [2]

46. 9702/22/F/M/22/Q2

Water leaves the end of a hose pipe at point P with a horizontal velocity of 6.6 m s^{-1} , as shown in Fig. 2.1.

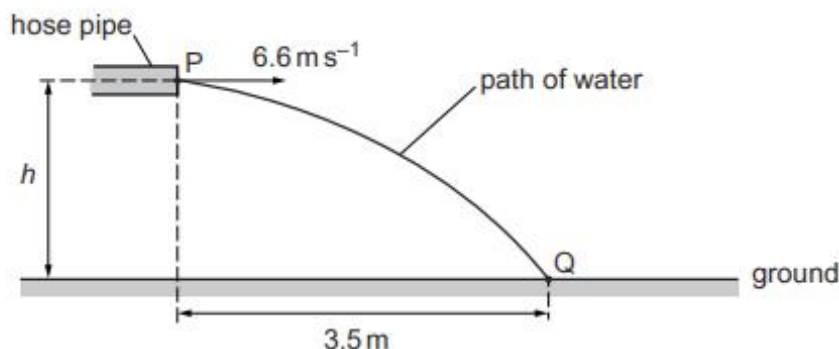


Fig. 2.1 (not to scale)

Point P is at height h above the ground. The water hits the ground at point Q. The horizontal distance from P to Q is 3.5 m.

Air resistance is negligible. Assume that the water between P and Q consists of non-interacting droplets of water and that the only force acting on each droplet is its weight.

- (a) Explain, briefly, why the horizontal component of the velocity of a droplet of water remains constant as it moves from P to Q.

.....
 [1]

- (b) Show that the time taken for a droplet of water to move from P to Q is 0.53 s.

[1]

- (c) Calculate height h .

$h =$ m [2]

- (d) For the movement of a droplet of water from P to Q, state and explain whether the displacement of the droplet is less than, more than or the same as the distance along its path.

.....
.....
..... [1]

- (e) Calculate the magnitude of the displacement of a droplet of water that moves from P to Q.

displacement = m [2]

[Total: 7]

47. 9702/22/M/J/22/Q3

A man standing on a wall throws a small ball vertically upwards with a velocity of 5.6 ms^{-1} . The ball leaves his hand when it is at a height of 3.1 m above the ground, as shown in Fig. 3.1.

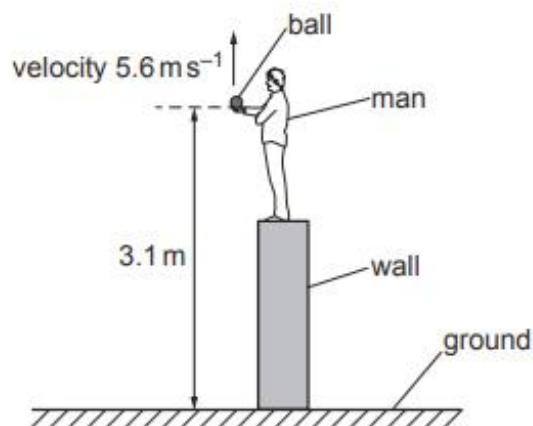


Fig. 3.1 (not to scale)

Assume that air resistance is negligible.

- (a) Show that the ball reaches a maximum height above the ground of 4.7 m .

[2]

- (b) The man does not catch the ball as it falls.

Calculate the time taken for the ball to fall from its maximum height to the ground.

time taken = s [2]

- (c) The ball leaves the man's hand at time $t = 0$ and hits the ground at time $t = T$.

On Fig. 3.2, sketch a graph to show the variation of the velocity v of the ball with time t from $t = 0$ to $t = T$. Numerical values of v and t are not required. Assume that v is positive in the upward direction.

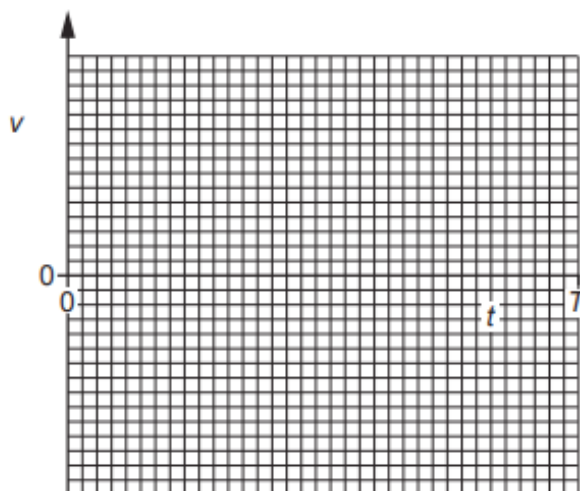


Fig. 3.2

[3]

- (d) State what is represented by the gradient of the graph in (c).

..... [1]

- (e) The man now throws a second ball with the same velocity and from the same height as the first ball. The mass of the second ball is greater than that of the first ball. Assume that air resistance is still negligible.

For the first and second balls, compare:

- (i) the magnitudes of their accelerations

..... [1]

- (ii) the speeds with which they hit the ground.

..... [1]

[Total: 10]

48.9702/23/M/J/22/Q2

An archer releases an arrow towards a target at a velocity of 65.0 m s^{-1} at an angle of 4.30° above the horizontal, as shown in Fig. 2.1.

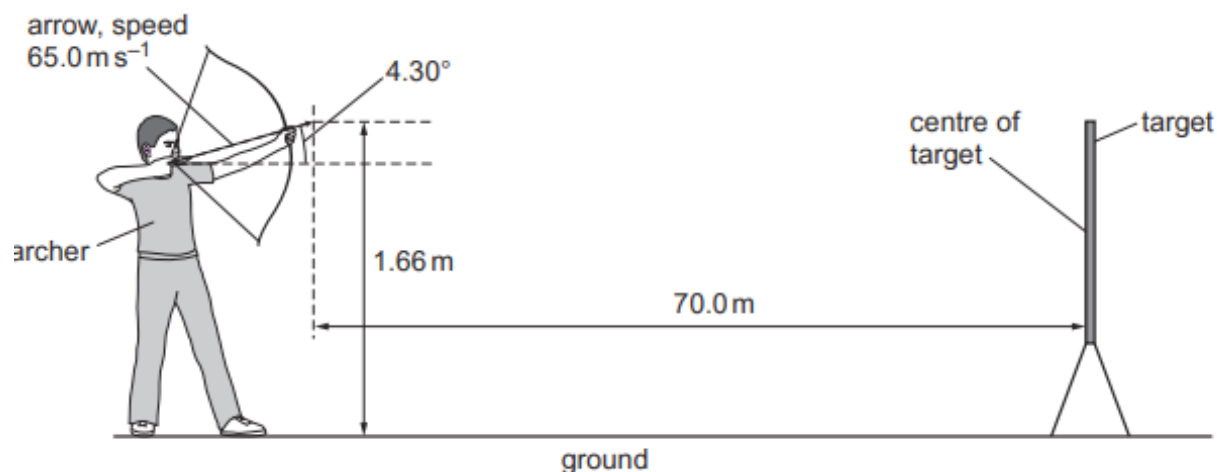


Fig. 2.1 (not to scale)

When released, the tip of the arrow is a horizontal distance of 70.0 m from the target and 1.66 m above the horizontal ground.

The arrow hits the centre of the target.

Assume that air resistance is negligible and that all the mass of the arrow is at its tip.

(a) Show that the time taken for the arrow to reach the target is 1.08 s .

[2]

(b) Calculate the height of the centre of the target above the ground.

height above ground = m [3]

- (c) By considering energy changes, state and explain how the final kinetic energy of the arrow as it hits the target compares with its initial kinetic energy immediately after release. A numerical calculation is not required.

.....

.....

.....

..... [2]

[Total: 7]

49. 9702/23/O/N/22/Q2

The engine of a toy rocket pushes gases vertically downwards and this results in the rocket accelerating vertically upwards from the ground.

The rocket starts to move from rest at time $t = 0$. The variation with time t of the vertical velocity v of the rocket for the first 0.30 s of the flight is shown in Fig. 2.1.

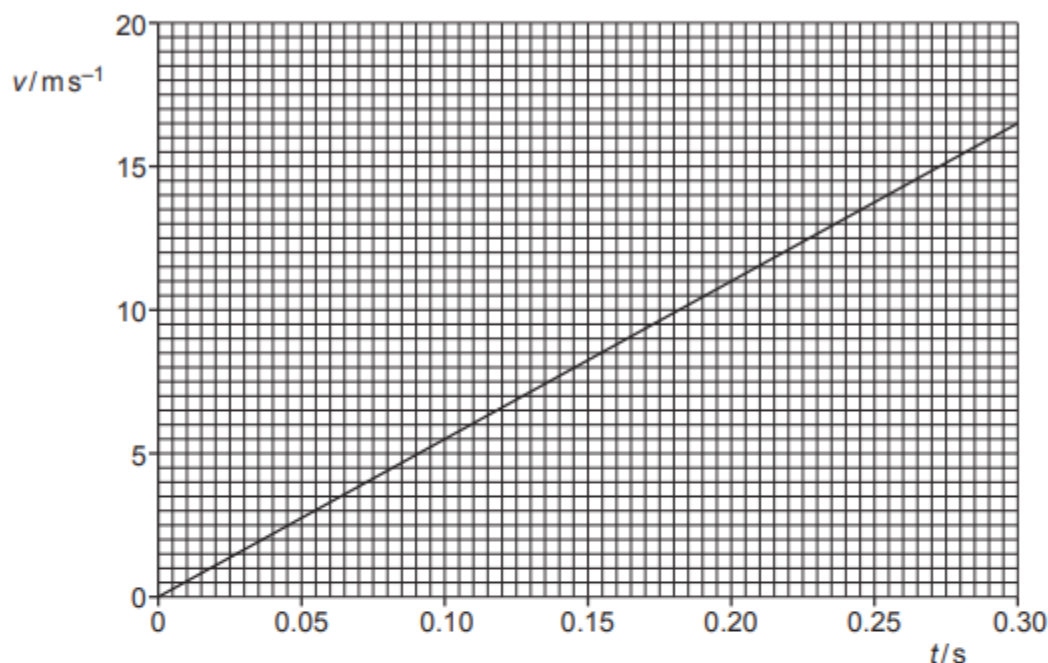


Fig. 2.1

As the rocket moves, the thrust force T provided by the rocket engine is 16 N. Assume that the mass of the rocket is constant for this part of its flight.

Assume that air resistance is negligible.

(a) For this part of the rocket's flight:

- (i) show that the acceleration of the rocket is 55 ms^{-2}

[1]

- (ii) state an expression for the resultant force F experienced by the rocket in terms of the thrust force T and the weight W of the rocket

[1]

(iii) calculate the mass of the rocket.

mass = kg [2]

(b) At time $t = 0.30$ s, a small piece of metal separates from the rocket.

Calculate:

(i) the height of the rocket above the ground at $t = 0.30$ s

height = m [2]

(ii) the speed at which the piece of metal strikes the ground.

speed = ms^{-1} [3]

[Total: 9]

50. 9702/21/O/N/23/Q3

A trolley A moves along a horizontal surface at a constant velocity towards another trolley B which is moving at a lower constant speed in the same direction. Fig. 3.1 shows the trolleys at time $t = 0$.



Fig. 3.1

Table 3.1 shows data for the trolleys.

Table 3.1

trolley	mass/kg	initial speed/ m s^{-1}
A	0.25	0.48
B	0.75	0.12

The two trolleys collide elastically and then separate. Resistive forces are negligible.

Fig. 3.2 shows the variation with time t of the velocity v for trolley B.

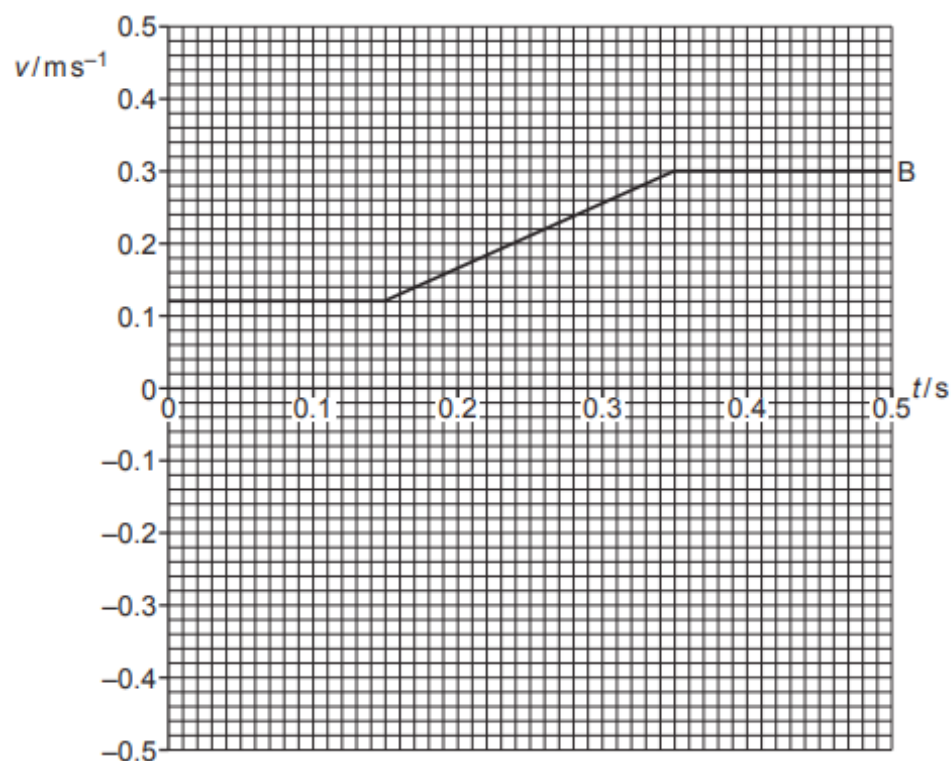


Fig. 3.2

(a) State what is represented by the area under a velocity–time graph.

..... [1]

(b) Use Table 3.1 and Fig. 3.2 to determine:

(i) the acceleration of trolley B during the collision

acceleration of B = ms^{-2} [2]

(ii) the magnitude and direction of the final velocity of trolley A.

magnitude = ms^{-1}

direction [3]

(c) On Fig. 3.2, sketch the variation of the velocity of trolley A with time t from $t = 0$ to $t = 0.50$ s. [3]

[Total: 9]

51. 9702/22/O/N/23/Q1

(a) In the following list, underline **all** quantities that are SI base quantities.

charge

electric current

force

time

[1]

(b) Under certain conditions, the distance s moved in a straight line by an object in time t is given by

$$s = \frac{1}{2}at^2$$

where a is the acceleration of the object.

State **two** conditions under which the above expression applies to the motion of the object.

1

2

[2]

(c) The variation with time t of the velocity v of a car that is moving in a straight line is shown in Fig. 1.1.

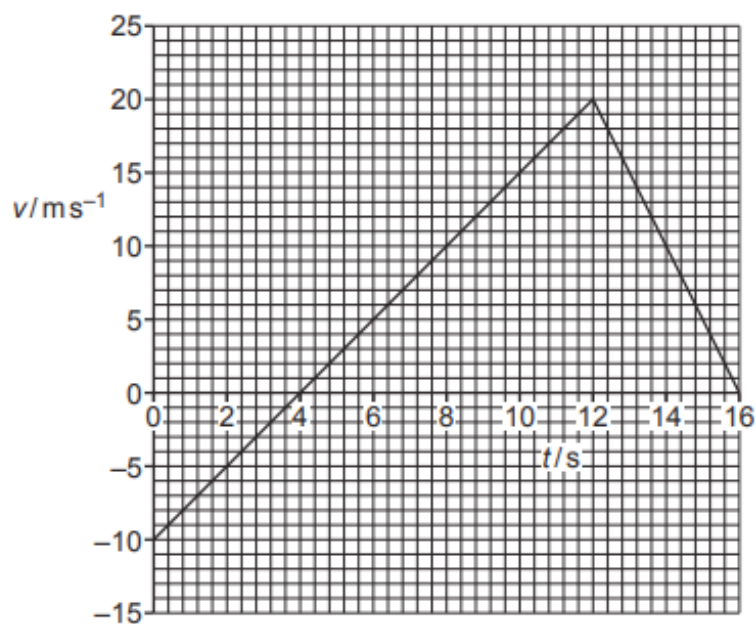


Fig. 1.1

- (i) Compare, qualitatively, the acceleration of the car at time $t = 8.0$ s and at time $t = 14.0$ s in terms of:

- magnitude

.....
.....

- direction.

.....
.....

[2]

- (ii) Determine the magnitude of the acceleration of the car at time $t = 4.0$ s.

acceleration = ms^{-2} [2]

- (iii) The car is at point X at time $t = 0$.

Determine the magnitude of the displacement of the car from X at time $t = 12.0$ s.

displacement = m [2]

[Total: 9]

52. 9702/23/O/N/23/Q2

A ball on horizontal ground is kicked towards a vertical wall. Fig. 2.1 shows the path of the ball.

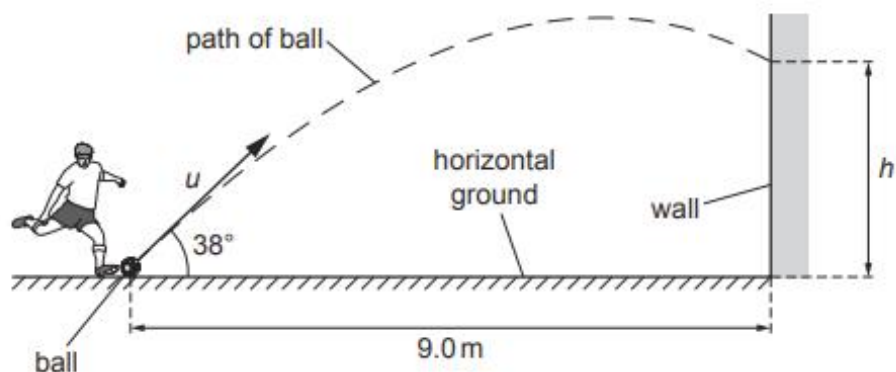


Fig. 2.1 (not to scale)

The ball has an initial velocity u at an angle of 38° to the ground. The ball travels a horizontal distance of 9.0 m before striking the wall at a height h above the ground. The horizontal component u_H of the initial velocity of the ball is 9.5 m s^{-1} .

Air resistance is negligible.

- (a) (i) Show that the time t for the ball to reach the wall is 0.95 s.

[1]

- (ii) Calculate the vertical component u_V of the initial velocity of the ball.

$$u_V = \dots \text{ ms}^{-1} \quad [2]$$

- (iii) Determine h .

$$h = \dots \text{ m} \quad [2]$$

- (b) The speed of the ball just after striking the wall is less than its speed just before striking the wall.

State what this indicates about the nature of the collision of the ball with the wall.

.....

..... [1]

[Total: 6]